

ASYMMETRIC INFORMATION, RELATIONAL
CONTRACTS AND PRICES:
EVIDENCE FROM AN EXPERIMENT WITH POTATO FARMERS IN PAKISTAN

Online Appendix: Pre-Specified Analysis*

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*The trial is registered at AEA RCT registry ID # AEARCTR-0012787. This appendix reproduces the pre-specified analysis plan for the registered trial and accompanies the current working paper based on the same randomized controlled trial. The title of the accompanying paper may differ from the registered title.

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1 Departures from PreSpecified Analysis

The registry entry (AEARCTR-0012787) diverges from the detailed pre analysis plan that was uploaded later in three respects. First, it states an expected sample of 2,100 farmers; this was an ex-ante estimate from the initial listing exercise, not a confirmed target, and was superseded by the baseline screening and consent process described in the main paper (991 farmers, 98 villages, in the final analysis sample). Second, it describes randomisation at the village level; we instead randomised, and cluster standard errors, at the patwar-circle (PC) level, the smallest administrative unit above the village, to avoid spillovers between neighbouring villages. Third, it lists price, sales, and network as primary outcomes; the network outcome was dropped after baseline piloting made clear that farmers were unable to report identifying information about the potential chain of intermediaries they sell to. A detailed outcome list was pre-registered instead in the accompanying pre-analysis plan, on which this Online Appendix is based.

The pre-specified outcome variable measuring the proportion of sample markets in which farmers sold potatoes (sale_markets) was excluded from the analysis due to insufficient variation. All farmers who reported their market for potato sales sold in only one market, and 96% sold their potatoes in one of the 17 sample markets, leaving little variation for statistical analysis.

2 Main Analysis

2.1 Primary Outcomes

Table 1: ITT Estimates: Sales, Quantities, Prices and Profits (2024)

[illegible]

Note: This table presents intent-to-treat estimates for sales, quantities, prices and profits for the pooled and disaggregated treatment estimations. "ln(Sales_2024)" is the log of total sales value received by the farmer in PKR per hectare of total cultivated area. "ln(Quantity_2024)" is the log of total quantity sold by the farmer in KGs per hectare of total cultivated area. "ln(Costs_2024)" is the log of total costs of the farmer in PKR per hectare of total cultivated area. "ln(Price_2024)" is the log of the total price received by the farmer in PKR/KG. "ln(Profit_2024)" is the log of the profit of the farmer from the 2024 season in PKR per hectare of total cultivated area. Each regression controls for whether the farmer sold potatoes in 2023, as well as the corresponding 2023 values of log of total sales value, log of total quantity sold, log of total costs, log of total price received, and log of profits. Strata fixed effects were used. Standard errors are clustered at the patwar-circle (PC) level throughout this appendix (77 PCs among the 991 farmers in the main analysis sample).

Table 2: ITT Estimates: Sales, Quantities, Prices and Profits (2025)

	(1) ln(Sales_2025)	(2) ln(Sales_2025)	(3) ln(Quantity_2025)	(4) ln(Quantity_2025)	(5) ln(Price_2025)	(6) ln(Price_2025)
Treated	0.098 (0.162)		0.089 (0.175)		-0.001 (0.048)	
Treatment 1		0.018 (0.168)		-0.001 (0.193)		0.011 (0.058)
Treatment 2		0.176 (0.182)		0.178 (0.200)		-0.014 (0.064)
Observations	322	322	322	322	322	322

Note: This table presents intent-to-treat estimates for sales, quantities, prices and profits for the pooled and disaggregated treatment estimations. "ln(Sales_2025)" is the log of total sales value received by the farmer in PKR per hectare of total cultivated area. "ln(Quantity_2025)" is the log of total quantity sold by the farmer in KGs per hectare of total cultivated area. "ln(Price_2025)" is the log of the total price received by the farmer in PKR/KG. Each regression controls for the corresponding 2024 values of log of total sales value, log of total quantity sold, and log of total price received. Strata fixed effects were used. Standard errors are clustered at the patwar-circle (PC) level throughout this appendix.

Table 3: LATE Estimates: Sales, Quantities, Prices and Profits (2024)

	(1) ln(Sales_2024)	(2) ln(Sales_2024)	(3) ln(Quantity_2024)	(4) ln(Quantity_2024)	(5) ln(Costs_2024)	(6) ln(Costs_2024)	(7) ln(Price_2024)	(8) ln(Price_2024)	(9) ln(Profit_2024)	(10) ln(Profit_2024)
Adopted Treatment	-0.001 (0.125)		0.008 (0.113)		-0.088 (0.110)		-0.011 (0.037)		0.026 (0.153)	
Adopted T1		-0.011 (0.134)		-0.024 (0.117)		-0.165 (0.114)		0.006 (0.044)		-0.004 (0.155)
Adopted T2		0.010 (0.142)		0.041 (0.123)		-0.008 (0.119)		-0.029 (0.042)		0.058 (0.202)
Observations	991	991	991	991	991	991	991	991	991	991

Note: ln(Sales), ln(Quantity), ln(Costs), ln(Price) and ln(Profit) are the natural logs of total sales value, quantity sold, cultivation/harvest costs, unit price received, and profit, each measured per hectare of total cultivated area. Adopted Treatment / Adopted T1 / Adopted T2 report the 2SLS Local Average Treatment Effect (LATE) of app adoption on the outcome, with adoption instrumented by the corresponding treatment assignment indicator. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.10.

Table 4: LATE Estimates: Sales, Quantities, Prices and Profits (2025)

	(1) ln(Sales_2025)	(2) ln(Sales_2025)	(3) ln(Quantity_2025)	(4) ln(Quantity_2025)	(5) ln(Price_2025)	(6) ln(Price_2025)
Adopted Treatment	0.098 (0.162)		0.089 (0.175)		-0.001 (0.048)	
Adopted T1		0.018 (0.168)		-0.001 (0.193)		0.011 (0.058)
Adopted T2		0.176 (0.182)		0.178 (0.200)		-0.014 (0.064)
Observations	322	322	322	322	322	322

Note: ln(Sales), ln(Quantity), ln(Costs), ln(Price) and ln(Profit) are the natural logs of total sales value, quantity sold, cultivation/harvest costs, unit price received, and profit, each measured per hectare of total cultivated area. Adopted Treatment / Adopted T1 / Adopted T2 report the 2SLS Local Average Treatment Effect (LATE) of app adoption on the outcome, with adoption instrumented by the corresponding treatment assignment indicator. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 5: ITT Estimates: Percentage of Sales by Buyer Type (2024)

	(1)	(2)	(3)	(4)	(5)	(6)
	CA 2024	CA 2024	Informal Agent 2024	Informal Agent 2024	Other 2024	Other 2024
Treated	-0.023 (0.063)		0.036 (0.060)		-0.018 (0.018)	
Treatment 1		-0.059 (0.059)		0.064 (0.058)		-0.009 (0.021)
Treatment 2		0.015 (0.087)		0.006 (0.083)		-0.027 (0.021)
Observations	991	991	991	991	991	991

Note: Outcomes are the share of the farmer’s total sales value sold to the Commission Agent (CA), to an informal agent, and to other buyer types. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.10.

Table 6: ITT Estimates: Percentage of Sales by Buyer Type (2025)

	(1) perc_CA_2025	(2) perc_CA_2025	(3) perc_infagent_2025	(4) perc_infagent_2025	(5) perc_other_2025	(6) perc_other_2025
Treated	0.121** (0.059)		-0.127** (0.064)		-0.002 (0.015)	
Treatment 1		0.060 (0.067)		-0.067 (0.070)		0.004 (0.023)
Treatment 2		0.179** (0.076)		-0.182** (0.080)		-0.009 (0.009)
Observations	322	322	322	322	322	322

Note: Outcomes are the share of the farmer's total sales value sold to the Commission Agent (CA), to an informal agent, and to other buyer types. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.10.

Table 7: LATE Estimates: Percentage of Sales by Buyer Type (2024)

	(1) CA 2024	(2) CA 2024	(3) Informal Agent 2024	(4) Informal Agent 2024	(5) Other 2024	(6) Other 2024
Adopted Treatment	-0.023 (0.063)		0.036 (0.060)		-0.018 (0.018)	
Adopted T1		-0.060 (0.059)		0.064 (0.058)		-0.009 (0.021)
Adopted T2		0.015 (0.087)		0.006 (0.083)		-0.027 (0.021)
Observations	991	991	991	991	991	991

Note: Outcomes are the share of the farmer's total sales value sold to the Commission Agent (CA), to an informal agent, and to other buyer types. Adopted Treatment / Adopted T1 / Adopted T2 report the 2SLS Local Average Treatment Effect (LATE) of app adoption on the outcome, with adoption instrumented by the corresponding treatment assignment indicator. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 8: LATE Estimates: Percentage of Sales by Buyer Type (2025)

	(1) perc_CA_2025	(2) perc_CA_2025	(3) perc_infagent_2025	(4) perc_infagent_2025	(5) perc_other_2025	(6) perc_other_2025
Adopted Treatment	0.121** (0.059)		-0.127** (0.064)		-0.002 (0.015)	
Adopted T1		0.060 (0.067)		-0.067 (0.070)		0.004 (0.023)
Adopted T2		0.179** (0.076)		-0.182** (0.080)		-0.009 (0.009)
Observations	322	322	322	322	322	322

Note: Outcomes are the share of the farmer's total sales value sold to the Commission Agent (CA), to an informal agent, and to other buyer types. Adopted Treatment / Adopted T1 / Adopted T2 report the 2SLS Local Average Treatment Effect (LATE) of app adoption on the outcome, with adoption instrumented by the corresponding treatment assignment indicator. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 9: ITT Estimates: Change of Buyer (2024)

	(1) Flexible	(2) Flexible	(3) New Agent	(4) New Agent	(5) Change Buyer	(6) Change Buyer	(7) Change Index	(8) Change Index
Treated	0.005 (0.052)		-0.174 (0.377)		-0.008 (0.068)		-0.001 (0.008)	
Treatment 1		0.025 (0.057)		0.073 (0.405)		0.090 (0.062)		0.014 (0.008)
Treatment 2		-0.015 (0.063)		-0.432 (0.440)		-0.111 (0.088)		-0.017 (0.011)
Observations	991	991	991	991	991	991	991	991

Note: Change Buyer is a binary indicator equal to one if the farmer sold to a different buyer/agent than at baseline. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 10: ITT Estimates: Change of Buyer (2025)

	(1) change_buyer_2025	(2) change_buyer_2025
Treated	0.080 (0.058)	
Treatment 1		0.105* (0.056)
Treatment 2		0.050 (0.080)
Observations	219	219

Note: Change Buyer is a binary indicator equal to one if the farmer sold to a different buyer/agent than at baseline. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 11: LATE Estimates: Change of Buyer (2024)

	(1) Flexible	(2) Flexible	(3) New Agent	(4) New Agent	(5) Change Buyer	(6) Change Index	(7) Change Index	(8) Change Buyer
Adopted Treatment	0.005 (0.053)		-0.174 (0.378)		-0.009 (0.068)	-0.001 (0.008)		
Adopted T1		0.025 (0.057)		0.074 (0.406)			0.014 (0.008)	0.091 (0.062)
Adopted T2		-0.015 (0.064)		-0.433 (0.441)			-0.017 (0.011)	-0.112 (0.088)
Observations	991	991	991	991	991	991	991	991

Note: Change Buyer is a binary indicator equal to one if the farmer sold to a different buyer/agent than at baseline. Adopted Treatment / Adopted T1 / Adopted T2 report the 2SLS Local Average Treatment Effect (LATE) of app adoption on the outcome, with adoption instrumented by the corresponding treatment assignment indicator. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.10.

Table 12: LATE Estimates: Change of Buyer (2025)

	(1) change_buyer_2025	(2) change_buyer_2025
Adopted Treatment	0.080 (0.058)	
Adopted T1		0.105* (0.056)
Adopted T2		0.050 (0.080)
Observations	219	219

Note: Change Buyer is a binary indicator equal to one if the farmer sold to a different buyer/agent than at baseline. Adopted Treatment / Adopted T1 / Adopted T2 report the 2SLS Local Average Treatment Effect (LATE) of app adoption on the outcome, with adoption instrumented by the corresponding treatment assignment indicator. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

2.2 Secondary Outcomes

Table 13: ITT Estimates: Harvest Costs

	(1)	(2)	(3)	(4)	(5)	(6)
	ln(Harvest Cost Farmer '24)	ln(Harvest Cost Farmer '24)	ln(Harvest Cost Buyer '24)	ln(Harvest Cost Buyer '24)	Harvest Cost Share '24	Harvest Cost Share '24
Treated	-0.190 (0.195)		0.198 (0.413)		-0.008 (0.024)	
Treatment 1		-0.374* (0.204)		0.878* (0.476)		-0.053** (0.025)
Treatment 2		-0.000 (0.256)		-0.509 (0.450)		0.039 (0.026)
Observations	991	991	991	991	991	991

Note: Outcomes are $\ln(\text{Harvest Cost, Farmer})$ and $\ln(\text{Harvest Cost, Buyer})$, $\ln(\text{cost} + 1)$ of harvest-time costs borne by the farmer and by the buyer respectively, and Harvest Cost Farmer's Share, the farmer-paid share of total harvest costs (farmer plus buyer). The $+1$ shift accommodates farmers with zero cost on that side of the transaction. Farmer-paid cost is zero for only 2.4% of the sample, but buyer-paid cost is zero for roughly 87%; columns (3)–(4) should be read with that large zero mass in mind, since a $\ln(x + 1)$ average blends it with a thin, highly skewed set of non-zero payers. Reported harvest-cost values were checked against reported prices and profits and corrected for flagged data-entry errors prior to any transformation. The main paper (Section 6.3) reports this margin instead as an extensive-margin (share of farmers with any buyer-paid cost) and intensive-margin (conditional amount paid) decomposition, which is not sensitive to this issue. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 14: LATE Estimates: Harvest Costs

	(1) ln(Harvest Cost Farmer '24)	(2) ln(Harvest Cost Farmer '24)	(3) ln(Harvest Cost Buyer '24)	(4) ln(Harvest Cost Buyer '24)	(5) Harvest Cost Share '24	(6) Harvest Cost Share '24
Adopted Treatment	-0.191 (0.196)		0.199 (0.414)		-0.008 (0.024)	
Adopted T1		-0.375* (0.204)		0.881* (0.478)		-0.053** (0.025)
Adopted T2		0.000 (0.257)		-0.511 (0.451)		0.039 (0.026)
Observations	991	991	991	991	991	991

Note: Outcomes are $\ln(\text{Harvest Cost, Farmer})$ and $\ln(\text{Harvest Cost, Buyer})$, $\ln(\text{cost} + 1)$ of harvest-time costs borne by the farmer and by the buyer respectively, and Harvest Cost Farmer's Share, the farmer-paid share of total harvest costs (farmer plus buyer). The $+1$ shift accommodates farmers with zero cost on that side of the transaction. Farmer-paid cost is zero for only 2.4% of the sample, but buyer-paid cost is zero for roughly 87%; columns (3)–(4) should be read with that large zero mass in mind, since a $\ln(x + 1)$ average blends it with a thin, highly skewed set of non-zero payers. Reported harvest-cost values were checked against reported prices and profits and corrected for flagged data-entry errors prior to any transformation. The main paper (Section 6.3) reports this margin instead as an extensive-margin (share of farmers with any buyer-paid cost) and intensive-margin (conditional amount paid) decomposition, which is not sensitive to this issue. Adopted Treatment / Adopted T1 / Adopted T2 report the 2SLS Local Average Treatment Effect (LATE) of app adoption on the outcome, with adoption instrumented by the corresponding treatment assignment indicator. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 15: ITT Estimates: Additional Services Provided by Buyer

	(1) Additional Services 2024	(2) Additional Services 2024
Treated	0.014 (0.036)	
Treatment 1		0.083** (0.039)
Treatment 2		-0.057 (0.038)
Observations	991	991

Note: Additional Services is a binary indicator equal to one if the buyer provided additional services (e.g. financial assistance or inputs) at cultivation. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.10.

Table 16: LATE Estimates: Additional Services Provided by Buyer

	(1) Additional Services 2024	(2) Additional Services 2024
Adopted Treatment	0.014 (0.036)	
Adopted T1		0.084** (0.039)
Adopted T2		-0.057 (0.038)
Observations	991	991

Note: Additional Services is a binary indicator equal to one if the buyer provided additional services (e.g. financial assistance or inputs) at cultivation. Adopted Treatment / Adopted T1 / Adopted T2 report the 2SLS Local Average Treatment Effect (LATE) of app adoption on the outcome, with adoption instrumented by the corresponding treatment assignment indicator. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 17: ITT Estimates: Agents Considered and Buyer Search Time

	(1) know_agents_2024	(2) know_agents_2024	(3) days_sale_2024	(4) days_sale_2024
Treated	-0.135 (0.215)		-1.249 (1.340)	
Treatment 1		0.064 (0.223)		-0.387 (1.382)
Treatment 2		-0.342 (0.276)		-2.149 (1.517)
Observations	991	991	989	989

Note: Outcomes are the number of buyers/agents the farmer considered before selling, and the number of days taken to find a buyer and complete the sale. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 18: LATE Estimates: Agents Considered and Buyer Search Time

	(1) know_agents_2024	(2) know_agents_2024	(3) days_sale_2024	(4) days_sale_2024
Adopted Treatment	-0.135 (0.216)		-1.252 (1.342)	
Adopted T1		0.064 (0.224)		-0.387 (1.385)
Adopted T2		-0.343 (0.277)		-2.155 (1.519)
Observations	991	991	989	989

Note: Outcomes are the number of buyers/agents the farmer considered before selling, and the number of days taken to find a buyer and complete the sale. Adopted Treatment / Adopted T1 / Adopted T2 report the 2SLS Local Average Treatment Effect (LATE) of app adoption on the outcome, with adoption instrumented by the corresponding treatment assignment indicator. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.10.

Table 19: ITT Estimates: Sowing Choice, New Agent Consideration and Credit Constraint (2025)

	Plant Potato (Expectation 2025)		New Agent (Expectation 2025)		Credit Constraint (Expectation 2025)	
	(1)	(2)	(3)	(4)	(5)	(6)
Treated	-0.022 (0.038)		-0.001 (0.056)		0.076** (0.030)	
Treatment 1		-0.050 (0.047)		-0.069 (0.066)		0.114*** (0.030)
Treatment 2		0.006 (0.035)		0.070 (0.058)		0.037 (0.035)
Observations	991	991	990	990	991	991

Note: Outcomes are Plant Potato (Expectation 2025) (intention to cultivate potatoes again in 2025), New Agent (Expectation 2025) (intention to consider a new Commission Agent), and Credit Constraint (Expectation 2025) (self-reported credit constraint), all reported at the 2024 endline, not realised 2025 outcomes. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 20: LATE Estimates: Sowing Choice, New Agent Consideration and Credit Constraint (2025)

	(1) plant_potato_2025	(2) plant_potato_2025	(3) new_agent_2025	(4) new_agent_2025	(5) credit_constraint_2025	(6) credit_constraint_2025
Adopted Treatment	-0.025 (0.037)		0.004 (0.056)		0.076** (0.030)	
Adopted T1		-0.053 (0.047)		-0.062 (0.066)		0.114*** (0.031)
Adopted T2		0.003 (0.034)		0.074 (0.058)		0.037 (0.034)
Observations	991	991	990	990	991	991

Note: Outcomes are Plant Potato (Expectation 2025) (intention to cultivate potatoes again in 2025), New Agent (Expectation 2025) (intention to consider a new Commission Agent), and Credit Constraint (Expectation 2025) (self-reported credit constraint), all reported at the 2024 endline, not realised 2025 outcomes. Adopted Treatment / Adopted T1 / Adopted T2 report the 2SLS Local Average Treatment Effect (LATE) of app adoption on the outcome, with adoption instrumented by the corresponding treatment assignment indicator. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

3 Heterogeneity Analysis

3.1 Education

Table 21: Heterogeneous Treatment Effects: Sales, Quantities, Prices and Profits, by education

	(1) ln(Sales_2024)	(2) ln(Sales_2024)	(3) ln(Quantity_2024)	(4) ln(Quantity_2024)	(5) ln(Costs_2024)	(6) ln(Costs_2024)	(7) ln(Price_2024)	(8) ln(Price_2024)	(9) ln(Profit_2024)	(10) ln(Profit_2024)
Treated	0.119 (0.162)		0.164 (0.122)		-0.027 (0.129)		-0.034 (0.043)		0.017 (0.214)	
High Education x Treated	-0.199 (0.193)		-0.251* (0.130)		-0.125 (0.148)		0.054 (0.057)		0.062 (0.234)	
Treatment 1		0.066 (0.168)		0.109 (0.126)		-0.135 (0.140)		-0.041 (0.053)		-0.036 (0.202)
Treatment 2		0.169 (0.195)		0.220 (0.135)		0.087 (0.133)		-0.029 (0.050)		0.072 (0.298)
High Education x T1		-0.102 (0.184)		-0.176 (0.135)		-0.064 (0.161)		0.106* (0.058)		0.118 (0.238)
High Education x T2		-0.301 (0.251)		-0.330** (0.155)		-0.195 (0.170)		0.002 (0.070)		0.001 (0.290)
Observations	976	976	976	976	976	976	976	976	976	976

Note: ln(Sales), ln(Quantity), ln(Costs), ln(Price) and ln(Profit) are the natural logs of total sales value, quantity sold, cultivation/harvest costs, unit price received, and profit, each measured per hectare of total cultivated area. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for baseline education (high vs. low), testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 22: Heterogeneous Treatment Effects: Percentage of Sales by Buyer Type, by education

	CA 2024		Informal Agent 2024		Other 2024	
	(1)	(2)	(3)	(4)	(5)	(6)
Treated	-0.018 (0.068)		0.022 (0.070)		-0.007 (0.025)	
Treatment 1		-0.058 (0.071)		0.051 (0.075)		0.006 (0.027)
Treatment 2		0.025 (0.083)		-0.009 (0.084)		-0.020 (0.029)
High Education x Treated	-0.008 (0.056)		0.030 (0.065)		-0.020 (0.032)	
High Education x T1		-0.009 (0.061)		0.026 (0.074)		-0.015 (0.039)
High Education x T2		-0.009 (0.072)		0.035 (0.079)		-0.024 (0.032)
Observations	976	976	976	976	976	976

Note: Outcomes are the share of the farmer's total sales value sold to the Commission Agent (CA), to an informal agent, and to other buyer types. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for baseline education (high vs. low), testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 23: Heterogeneous Treatment Effects: Flexible Arrangements, New Agent Consideration and Buyer Change, by education

	Flexible 2024		New Agent 2024		Change Buyer 2024	
	(1)	(2)	(3)	(4)	(5)	(6)
Treated	0.020 (0.062)		-0.228 (0.418)		-0.027 (0.073)	
Treatment 1		0.043 (0.067)		-0.146 (0.505)		0.056 (0.075)
Treatment 2		-0.005 (0.072)		-0.341 (0.441)		-0.120 (0.088)
High Education x Treated	-0.039 (0.041)		0.107 (0.361)		0.032 (0.071)	
High Education x T1		-0.053 (0.043)		0.458 (0.441)		0.064 (0.079)
High Education x T2		-0.024 (0.051)		-0.238 (0.383)		0.008 (0.082)
Observations	976	976	976	976	976	976

Note: Change Buyer is a binary indicator equal to one if the farmer sold to a different buyer/agent than at baseline. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for baseline education (high vs. low), testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 24: Heterogeneous Treatment Effects: Harvest Costs, by education

	(1) ln(Harvest Cost Farmer '24)	(2) ln(Harvest Cost Farmer '24)	(3) ln(Harvest Cost Buyer '24)	(4) ln(Harvest Cost Buyer '24)	(5) Harvest Cost Share '24	(6) Harvest Cost Share '24
Treated	-0.282 (0.278)		0.477 (0.436)		-0.037 (0.025)	
High Education x Treated	0.114 (0.311)		-0.589 (0.460)		0.056* (0.031)	
Treatment 1		-0.390 (0.308)		1.062** (0.507)		-0.073** (0.029)
Treatment 2		-0.154 (0.333)		-0.180 (0.471)		0.003 (0.027)
High Education x T1		-0.007 (0.353)		-0.323 (0.560)		0.038 (0.041)
High Education x T2		0.226 (0.336)		-0.807* (0.442)		0.070** (0.028)
Observations	976	976	976	976	976	976

Note: Outcomes are ln(Harvest Cost, Farmer), the log of harvest-time costs borne by the farmer, and Harvest Cost Farmer's Share, the farmer-paid share of total harvest costs (farmer plus buyer). Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for baseline education (high vs. low), testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.10.

Table 25: Heterogeneous Treatment Effects: Additional Services Provided by Buyer, by education

	Additional Services 2024	
	(1)	(2)
Treated	0.043 (0.038)	
Treatment 1		0.106** (0.045)
Treatment 2		-0.026 (0.040)
High Education x Treated	-0.058 (0.044)	
High Education x T1		-0.038 (0.053)
High Education x T2		-0.074* (0.043)
Observations	976	976

Note: Additional Services is a binary indicator equal to one if the buyer provided additional services (e.g. financial assistance or inputs) at cultivation. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for baseline education (high vs. low), testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 26: Heterogeneous Treatment Effects: Agents Considered and Buyer Search Time, by education

	Agents Considered 2024 (1)	(2)	Buyer Search Time (Days) 2024 (3)	(4)
Treated	-0.301* (0.182)		-2.884* (1.697)	
Treatment 1		-0.223 (0.218)		-2.079 (1.761)
Treatment 2		-0.391* (0.199)		-3.745** (1.754)
High Education x Treated	0.351** (0.152)		3.299* (1.731)	
High Education x T1		0.423** (0.168)		3.014 (2.049)
High Education x T2		0.287 (0.202)		3.653** (1.819)
Observations	976	976	974	974

Note: Outcomes are the number of buyers/agents the farmer considered before selling, and the number of days taken to find a buyer and complete the sale. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for baseline education (high vs. low), testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 27: Heterogeneous Treatment Effects: Sowing Choice, New Agent Consideration and Credit Constraint, by education

	Plant Potato (Expectation 2025) (1)		New Agent (Expectation 2025) (3)		Credit Constraint (Expectation 2025) (5)	
	(2)		(4)		(6)	
Treated	-0.034 (0.038)		0.009 (0.062)		0.075** (0.035)	
Treatment 1		-0.071 (0.047)		-0.023 (0.081)		0.108*** (0.037)
Treatment 2		0.006 (0.038)		0.047 (0.062)		0.039 (0.043)
High Education x Treated	0.016 (0.054)		-0.019 (0.067)		0.006 (0.037)	
High Education x T1		0.026 (0.068)		-0.061 (0.078)		0.014 (0.046)
High Education x T2		0.003 (0.063)		0.019 (0.069)		0.001 (0.049)
Observations	976	976	975	975	976	976

Note: Outcomes are Plant Potato (Expectation 2025) (intention to cultivate potatoes again in 2025), New Agent (Expectation 2025) (intention to consider a new Commission Agent), and Credit Constraint (Expectation 2025) (self-reported credit constraint), all reported at the 2024 endline, not realised 2025 outcomes. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for baseline education (high vs. low), testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

3.2 Numeracy

Table 28: Heterogeneous Treatment Effects: Sales, Quantities, Prices and Profits, by numeracy

	(1) ln(Sales_2024)	(2) ln(Sales_2024)	(3) ln(Quantity_2024)	(4) ln(Quantity_2024)	(5) ln(Costs_2024)	(6) ln(Costs_2024)	(7) ln(Price_2024)	(8) ln(Price_2024)	(9) ln(Profit_2024)	(10) ln(Profit_2024)
Treated	0.065 (0.124)		0.058 (0.094)		-0.084 (0.103)		0.017 (0.041)		0.088 (0.162)	
High Numeracy x Treated	-0.253 (0.166)		-0.100 (0.151)		0.004 (0.126)		-0.136* (0.072)		-0.260 (0.226)	
Treatment 1		0.058 (0.139)		0.043 (0.108)		-0.179 (0.110)		0.039 (0.046)		0.120 (0.155)
Treatment 2		0.071 (0.145)		0.074 (0.101)		0.010 (0.111)		-0.004 (0.048)		0.055 (0.220)
High Numeracy x T1		-0.214 (0.186)		-0.106 (0.153)		0.086 (0.116)		-0.138* (0.083)		-0.540* (0.311)
High Numeracy x T2		-0.308 (0.252)		-0.084 (0.238)		-0.079 (0.215)		-0.142* (0.085)		0.135 (0.278)
Observations	977	977	977	977	977	977	977	977	977	977

Note: ln(Sales), ln(Quantity), ln(Costs), ln(Price) and ln(Profit) are the natural logs of total sales value, quantity sold, cultivation/harvest costs, unit price received, and profit, each measured per hectare of total cultivated area. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for baseline numeracy, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.10.

Table 29: Heterogeneous Treatment Effects: Percentage of Sales by Buyer Type, by numeracy

	CA 2024		Informal Agent 2024		Other 2024	
	(1)	(2)	(3)	(4)	(5)	(6)
Treated	-0.010 (0.060)		0.016 (0.058)		-0.006 (0.015)	
Treatment 1		-0.051 (0.054)		0.043 (0.055)		0.010 (0.019)
Treatment 2		0.031 (0.085)		-0.011 (0.083)		-0.021 (0.016)
High Numeracy x Treated	-0.048 (0.069)		0.094 (0.070)		-0.058 (0.059)	
High Numeracy x T1		-0.055 (0.068)		0.100 (0.073)		-0.061 (0.060)
High Numeracy x T2		-0.024 (0.100)		0.075 (0.094)		-0.060 (0.060)
Observations	977	977	977	977	977	977

Note: Outcomes are the share of the farmer's total sales value sold to the Commission Agent (CA), to an informal agent, and to other buyer types. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for baseline numeracy, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.10.

Table 30: Heterogeneous Treatment Effects: Flexible Arrangements, New Agent Consideration and Buyer Change, by numeracy

	Flexible 2024 (1)	(2)	New Agent 2024 (3)	(4)	Change Buyer 2024 (5)	(6)
Treated	0.028 (0.055)		-0.261 (0.391)		-0.059 (0.070)	
Treatment 1		0.047 (0.056)		-0.001 (0.446)		0.050 (0.064)
Treatment 2		0.009 (0.067)		-0.521 (0.439)		-0.168** (0.085)
High Numeracy x Treated	-0.157** (0.067)		0.517 (0.506)		0.256*** (0.073)	
High Numeracy x T1		-0.158* (0.089)		0.486 (0.619)		0.190** (0.084)
High Numeracy x T2		-0.163** (0.064)		0.456 (0.479)		0.306*** (0.093)
Observations	977	977	977	977	977	977

Note: Change Buyer is a binary indicator equal to one if the farmer sold to a different buyer/agent than at baseline. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for baseline numeracy, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 31: Heterogeneous Treatment Effects: Harvest Costs, by numeracy

	(1) ln(Harvest Cost Farmer '24)	(2) ln(Harvest Cost Farmer '24)	(3) ln(Harvest Cost Buyer '24)	(4) ln(Harvest Cost Buyer '24)	(5) Harvest Cost Share '24	(6) Harvest Cost Share '24
Treated	-0.172 (0.181)		0.244 (0.448)		-0.020 (0.025)	
High Numeracy x Treated	-0.232 (0.303)		-0.408 (0.583)		0.063* (0.034)	
Treatment 1		-0.419** (0.193)		1.076** (0.539)		-0.072*** (0.027)
Treatment 2		0.078 (0.250)		-0.588 (0.478)		0.033 (0.028)
High Numeracy x T1		0.149 (0.331)		-0.911 (0.778)		0.095** (0.046)
High Numeracy x T2		-0.685** (0.315)		-0.013 (0.543)		0.037 (0.029)
Observations	977	977	977	977	977	977

Note: Outcomes are ln(Harvest Cost, Farmer), the log of harvest-time costs borne by the farmer, and Harvest Cost Farmer's Share, the farmer-paid share of total harvest costs (farmer plus buyer). Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for baseline numeracy, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.10.

Table 32: Heterogeneous Treatment Effects: Additional Services Provided by Buyer, by numeracy

	Additional Services 2024	
	(1)	(2)
Treated	0.014 (0.040)	
Treatment 1		0.102** (0.044)
Treatment 2		-0.074* (0.042)
High Numeracy x Treated	-0.003 (0.059)	
High Numeracy x T1		-0.079 (0.068)
High Numeracy x T2		0.073 (0.064)
Observations	977	977

Note: Additional Services is a binary indicator equal to one if the buyer provided additional services (e.g. financial assistance or inputs) at cultivation. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for baseline numeracy, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 33: Heterogeneous Treatment Effects: Agents Considered and Buyer Search Time, by numeracy

	Agents Considered 2024		Buyer Search Time (Days) 2024	
	(1)	(2)	(3)	(4)
Treated	-0.230 (0.180)		-2.216 (1.461)	
Treatment 1		-0.131 (0.199)		-1.874 (1.481)
Treatment 2		-0.328 (0.226)		-2.555 (1.642)
High Numeracy x Treated	0.570*** (0.193)		5.291** (2.267)	
High Numeracy x T1		0.607*** (0.219)		6.679** (2.766)
High Numeracy x T2		0.475** (0.200)		3.122 (2.127)
Observations	977	977	975	975

Note: Outcomes are the number of buyers/agents the farmer considered before selling, and the number of days taken to find a buyer and complete the sale. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for baseline numeracy, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 34: Heterogeneous Treatment Effects: Sowing Choice, New Agent Consideration and Credit Constraint, by numeracy

	Plant Potato (Expectation 2025)		New Agent (Expectation 2025)		Credit Constraint (Expectation 2025)	
	(1)	(2)	(3)	(4)	(5)	(6)
Treated	-0.003 (0.037)		-0.001 (0.048)		0.078** (0.034)	
Treatment 1		-0.041 (0.046)		-0.051 (0.059)		0.115*** (0.036)
Treatment 2		0.035 (0.036)		0.050 (0.051)		0.042 (0.039)
High Numeracy x Treated	-0.131* (0.067)		0.011 (0.075)		-0.002 (0.044)	
High Numeracy x T1		-0.092 (0.074)		-0.005 (0.078)		0.000 (0.057)
High Numeracy x T2		-0.172* (0.093)		0.054 (0.089)		-0.021 (0.044)
Observations	977	977	976	976	977	977

Note: Outcomes are Plant Potato (Expectation 2025) (intention to cultivate potatoes again in 2025), New Agent (Expectation 2025) (intention to consider a new Commission Agent), and Credit Constraint (Expectation 2025) (self-reported credit constraint), all reported at the 2024 endline, not realised 2025 outcomes. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for baseline numeracy, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.10.

3.3 Cultivated Area

Table 35: Heterogeneous Treatment Effects: Sales, Quantities, Prices and Profits, by cultivated area

	(1) ln(Sales_2024)	(2) ln(Sales_2024)	(3) ln(Quantity_2024)	(4) ln(Quantity_2024)	(5) ln(Costs_2024)	(6) ln(Costs_2024)	(7) ln(Price_2024)	(8) ln(Price_2024)	(9) ln(Profit_2024)	(10) ln(Profit_2024)
Treated	-0.023 (0.172)		-0.085 (0.131)		-0.172 (0.119)		0.063 (0.053)		-0.017 (0.115)	
High Cultivated Area x Treated	0.071 (0.203)		0.243* (0.138)		0.161 (0.120)		-0.144* (0.076)		0.099 (0.222)	
Treatment 1		-0.075 (0.180)		-0.179 (0.141)		-0.245* (0.142)		0.090 (0.062)		-0.107 (0.126)
Treatment 2		0.020 (0.191)		-0.003 (0.143)		-0.105 (0.134)		0.038 (0.059)		0.062 (0.127)
High Cultivated Area x T1		0.188 (0.192)		0.405** (0.158)		0.185 (0.158)		-0.156** (0.078)		0.254 (0.204)
High Cultivated Area x T2		-0.058 (0.272)		0.072 (0.164)		0.151 (0.139)		-0.136 (0.095)		-0.063 (0.335)
Observations	977	977	977	977	977	977	977	977	977	977

Note: ln(Sales), ln(Quantity), ln(Costs), ln(Price) and ln(Profit) are the natural logs of total sales value, quantity sold, cultivation/harvest costs, unit price received, and profit, each measured per hectare of total cultivated area. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for baseline cultivated area, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.10.

Table 36: Heterogeneous Treatment Effects: Percentage of Sales by Buyer Type, by cultivated area

	CA 2024		Informal Agent 2024		Other 2024	
	(1)	(2)	(3)	(4)	(5)	(6)
Treated	-0.013 (0.071)		0.003 (0.069)		0.008 (0.017)	
Treatment 1		-0.095 (0.064)		0.065 (0.066)		0.028 (0.025)
Treatment 2		0.062 (0.095)		-0.054 (0.095)		-0.011 (0.015)
High Cultivated Area x Treated	-0.018 (0.068)		0.067 (0.069)		-0.049* (0.029)	
High Cultivated Area x T1		0.062 (0.074)		-0.002 (0.081)		-0.060* (0.034)
High Cultivated Area x T2		-0.093 (0.075)		0.134* (0.074)		-0.042 (0.032)
Observations	977	977	977	977	977	977

Note: Outcomes are the share of the farmer's total sales value sold to the Commission Agent (CA), to an informal agent, and to other buyer types. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for baseline cultivated area, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.10.

Table 37: Heterogeneous Treatment Effects: Flexible Arrangements, New Agent Consideration and Buyer Change, by cultivated area

	Flexible 2024 (1)	(2)	New Agent 2024 (3)	(4)	Change Buyer 2024 (5)	(6)
Treated	-0.010 (0.056)		0.129 (0.436)		0.001 (0.068)	
Treatment 1		0.029 (0.059)		0.388 (0.555)		0.134** (0.061)
Treatment 2		-0.045 (0.066)		-0.116 (0.488)		-0.121 (0.088)
High Cultivated Area x Treated	0.021 (0.035)		-0.620 (0.431)		-0.027 (0.069)	
High Cultivated Area x T1		-0.022 (0.037)		-0.597 (0.625)		-0.093 (0.074)
High Cultivated Area x T2		0.063 (0.048)		-0.717* (0.388)		0.018 (0.073)
Observations	977	977	977	977	977	977

Note: Change Buyer is a binary indicator equal to one if the farmer sold to a different buyer/agent than at baseline. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for baseline cultivated area, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 38: Heterogeneous Treatment Effects: Harvest Costs, by cultivated area

	(1)	(2)	(3)	(4)	(5)	(6)
	ln(Harvest Cost Farmer '24)	ln(Harvest Cost Farmer '24)	ln(Harvest Cost Buyer '24)	ln(Harvest Cost Buyer '24)	Harvest Cost Share '24	Harvest Cost Share '24
Treated	-0.203 (0.243)		0.095 (0.463)		0.003 (0.032)	
High Cultivated Area x Treated	0.036 (0.276)		0.197 (0.464)		-0.024 (0.030)	
Treatment 1		-0.371 (0.286)		0.840 (0.584)		-0.040 (0.031)
Treatment 2		-0.043 (0.316)		-0.604 (0.481)		0.043 (0.038)
High Cultivated Area x T1		-0.023 (0.408)		0.134 (0.635)		-0.030 (0.035)
High Cultivated Area x T2		0.153 (0.278)		0.077 (0.476)		-0.007 (0.035)
Observations	977	977	977	977	977	977

Note: Outcomes are ln(Harvest Cost, Farmer), the log of harvest-time costs borne by the farmer, and Harvest Cost Farmer’s Share, the farmer-paid share of total harvest costs (farmer plus buyer). Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for baseline cultivated area, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.10.

Table 39: Heterogeneous Treatment Effects: Additional Services Provided by Buyer, by cultivated area

	Additional Services 2024	
	(1)	(2)
Treated	0.032 (0.043)	
Treatment 1		0.123** (0.050)
Treatment 2		-0.052 (0.043)
High Cultivated Area x Treated	-0.035 (0.044)	
High Cultivated Area x T1		-0.070 (0.057)
High Cultivated Area x T2		-0.017 (0.044)
Observations	977	977

Note: Additional Services is a binary indicator equal to one if the buyer provided additional services (e.g. financial assistance or inputs) at cultivation. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for baseline cultivated area, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 40: Heterogeneous Treatment Effects: Agents Considered and Buyer Search Time, by cultivated area

	Agents Considered 2024		Buyer Search Time (Days) 2024	
	(1)	(2)	(3)	(4)
Treated	-0.264 (0.199)		-0.834 (1.811)	
Treatment 1		-0.126 (0.219)		0.236 (1.969)
Treatment 2		-0.391 (0.244)		-1.802 (1.943)
High Cultivated Area x Treated	0.282 (0.191)		-0.810 (2.428)	
High Cultivated Area x T1		0.222 (0.212)		-1.701 (2.651)
High Cultivated Area x T2		0.318 (0.212)		-0.025 (2.491)
Observations	977	977	975	975

Note: Outcomes are the number of buyers/agents the farmer considered before selling, and the number of days taken to find a buyer and complete the sale. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for baseline cultivated area, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 41: Heterogeneous Treatment Effects: Sowing Choice, New Agent Consideration and Credit Constraint, by cultivated area

	Plant Potato (Expectation 2025)		New Agent (Expectation 2025)		Credit Constraint (Expectation 2025)
	(1)	(2)	(3)	(4)	(5)
	(6)				
Treated	-0.057 (0.043)		-0.001 (0.058)		0.097*** (0.032)
Treatment 1		-0.090* (0.051)		-0.098 (0.064)	0.148*** (0.031)
Treatment 2		-0.025 (0.045)		0.087 (0.061)	0.049 (0.037)
High Cultivated Area x Treated	0.065 (0.055)		0.004 (0.071)		-0.038 (0.033)
High Cultivated Area x T1		0.065 (0.065)		0.086 (0.080)	-0.067** (0.033)
High Cultivated Area x T2		0.075 (0.058)		-0.070 (0.097)	-0.017 (0.054)
Observations	977	977	976	976	977

Note: Outcomes are Plant Potato (Expectation 2025) (intention to cultivate potatoes again in 2025), New Agent (Expectation 2025) (intention to consider a new Commission Agent), and Credit Constraint (Expectation 2025) (self-reported credit constraint), all reported at the 2024 endline, not realised 2025 outcomes. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for baseline cultivated area, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.10.

3.4 Agricultural Empowerment

Table 42: Heterogeneous Treatment Effects: Sales, Quantities, Prices and Profits, by agricultural empowerment

	(1) ln(Sales_2024)	(2) ln(Sales_2024)	(3) ln(Quantity_2024)	(4) ln(Quantity_2024)	(5) ln(Costs_2024)	(6) ln(Costs_2024)	(7) ln(Price_2024)	(8) ln(Price_2024)	(9) ln(Profit_2024)	(10) ln(Profit_2024)
Treated	0.167 (0.141)		0.155 (0.108)		-0.017 (0.105)		0.000 (0.043)		0.192 (0.178)	
High Agri Empowerment x Treated	-0.297 (0.199)		-0.228 (0.148)		-0.137 (0.149)		-0.016 (0.061)		-0.297 (0.231)	
Treatment 1		0.162 (0.150)		0.165 (0.108)		-0.096 (0.111)		0.003 (0.048)		0.139 (0.187)
Treatment 2		0.173 (0.174)		0.144 (0.143)		0.074 (0.126)		-0.004 (0.052)		0.252 (0.204)
High Agri Empowerment x T1		-0.299 (0.209)		-0.292* (0.152)		-0.134 (0.165)		0.018 (0.065)		-0.243 (0.234)
High Agri Empowerment x T2		-0.295 (0.231)		-0.167 (0.182)		-0.153 (0.166)		-0.048 (0.072)		-0.356 (0.282)
Observations	977	977	977	977	977	977	977	977	977	977

Note: ln(Sales), ln(Quantity), ln(Costs), ln(Price) and ln(Profit) are the natural logs of total sales value, quantity sold, cultivation/harvest costs, unit price received, and profit, each measured per hectare of total cultivated area. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for baseline agricultural empowerment, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.10.

Table 43: Heterogeneous Treatment Effects: Percentage of Sales by Buyer Type, by agricultural empowerment

	CA 2024		Informal Agent 2024		Other 2024	
	(1)	(2)	(3)	(4)	(5)	(6)
Treated	0.020 (0.071)		-0.010 (0.066)		-0.014 (0.019)	
Treatment 1		-0.050 (0.066)		0.041 (0.064)		0.006 (0.025)
Treatment 2		0.100 (0.096)		-0.067 (0.093)		-0.037** (0.016)
High Agri Empowerment x Treated	-0.085 (0.067)		0.093 (0.062)		-0.004 (0.032)	
High Agri Empowerment x T1		-0.024 (0.068)		0.043 (0.065)		-0.014 (0.039)
High Agri Empowerment x T2		-0.156** (0.071)		0.149** (0.067)		0.008 (0.033)
Observations	977	977	977	977	977	977

Note: Outcomes are the share of the farmer's total sales value sold to the Commission Agent (CA), to an informal agent, and to other buyer types. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for baseline agricultural empowerment, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.10.

Table 44: Heterogeneous Treatment Effects: Flexible Arrangements, New Agent Consideration and Buyer Change, by agricultural empowerment

	Flexible 2024 (1)	(2)	New Agent 2024 (3)	(4)	Change Buyer 2024 (5)	(6)
Treated	0.000 (0.060)		-0.279 (0.478)		0.019 (0.073)	
Treatment 1		0.002 (0.063)		0.094 (0.530)		0.125* (0.066)
Treatment 2		-0.002 (0.079)		-0.701 (0.544)		-0.100 (0.105)
High Agri Empowerment x Treated	-0.000 (0.053)		0.223 (0.435)		-0.062 (0.072)	
High Agri Empowerment x T1		0.030 (0.057)		-0.008 (0.509)		-0.075 (0.074)
High Agri Empowerment x T2		-0.029 (0.073)		0.500 (0.455)		-0.032 (0.084)
Observations	977	977	977	977	977	977

Note: Change Buyer is a binary indicator equal to one if the farmer sold to a different buyer/agent than at baseline. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for baseline agricultural empowerment, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.10.

Table 45: Heterogeneous Treatment Effects: Harvest Costs, by agricultural empowerment

	(1) ln(Harvest Cost Farmer '24)	(2) ln(Harvest Cost Farmer '24)	(3) ln(Harvest Cost Buyer '24)	(4) ln(Harvest Cost Buyer '24)	(5) Harvest Cost Share '24	(6) Harvest Cost Share '24
Treated	0.036 (0.210)		-0.110 (0.512)		0.006 (0.030)	
High Agri Empowerment x Treated	-0.535** (0.243)		0.588 (0.577)		-0.031 (0.030)	
Treatment 1		-0.020 (0.202)		0.775 (0.567)		-0.037 (0.033)
Treatment 2		0.102 (0.297)		-1.115** (0.483)		0.056* (0.031)
High Agri Empowerment x T1		-0.800*** (0.295)		0.267 (0.654)		-0.037 (0.034)
High Agri Empowerment x T2		-0.300 (0.240)		1.034 (0.650)		-0.033 (0.031)
Observations	977	977	977	977	977	977

Note: Outcomes are ln(Harvest Cost, Farmer), the log of harvest-time costs borne by the farmer, and Harvest Cost Farmer's Share, the farmer-paid share of total harvest costs (farmer plus buyer). Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for baseline agricultural empowerment, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.10.

Table 46: Heterogeneous Treatment Effects: Additional Services Provided by Buyer, by agricultural empowerment

	Additional Services 2024	
	(1)	(2)
Treated	-0.004 (0.050)	
Treatment 1		0.081 (0.054)
Treatment 2		-0.101** (0.047)
High Agri Empowerment x Treated	0.038 (0.059)	
High Agri Empowerment x T1		0.014 (0.061)
High Agri Empowerment x T2		0.075 (0.067)
Observations	977	977

Note: Additional Services is a binary indicator equal to one if the buyer provided additional services (e.g. financial assistance or inputs) at cultivation. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for baseline agricultural empowerment, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 47: Heterogeneous Treatment Effects: Agents Considered and Buyer Search Time, by agricultural empowerment

	Agents Considered 2024		Buyer Search Time (Days) 2024	
	(1)	(2)	(3)	(4)
Treated	-0.241 (0.254)		-1.860 (1.722)	
Treatment 1		-0.093 (0.264)		-0.883 (1.765)
Treatment 2		-0.408 (0.301)		-2.974 (2.083)
High Agri Empowerment x Treated	0.224 (0.241)		1.190 (1.607)	
High Agri Empowerment x T1		0.151 (0.246)		0.486 (1.650)
High Agri Empowerment x T2		0.317 (0.273)		2.023 (2.134)
Observations	977	977	975	975

Note: Outcomes are the number of buyers/agents the farmer considered before selling, and the number of days taken to find a buyer and complete the sale. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for baseline agricultural empowerment, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 48: Heterogeneous Treatment Effects: Sowing Choice, New Agent Consideration and Credit Constraint, by agricultural empowerment

	Plant Potato (Expectation 2025)		New Agent (Expectation 2025)		Credit Constraint (Expectation 2025)	
	(1)	(2)	(3)	(4)	(5)	(6)
Treated	-0.017 (0.040)		0.003 (0.056)		0.085*** (0.028)	
Treatment 1		-0.020 (0.046)		-0.022 (0.072)		0.124*** (0.032)
Treatment 2		-0.012 (0.045)		0.033 (0.060)		0.041 (0.029)
High Agri Empowerment x Treated	-0.018 (0.051)		-0.008 (0.069)		-0.014 (0.033)	
High Agri Empowerment x T1		-0.080 (0.065)		-0.067 (0.077)		-0.017 (0.046)
High Agri Empowerment x T2		0.039 (0.053)		0.043 (0.085)		-0.005 (0.039)
Observations	977	977	976	976	977	977

Note: Outcomes are Plant Potato (Expectation 2025) (intention to cultivate potatoes again in 2025), New Agent (Expectation 2025) (intention to consider a new Commission Agent), and Credit Constraint (Expectation 2025) (self-reported credit constraint), all reported at the 2024 endline, not realised 2025 outcomes. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for baseline agricultural empowerment, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

3.5 Household Empowerment

Table 49: Heterogeneous Treatment Effects: Sales, Quantities, Prices and Profits, by household empowerment

	(1) ln(Sales_2024)	(2) ln(Sales_2024)	(3) ln(Quantity_2024)	(4) ln(Quantity_2024)	(5) ln(Costs_2024)	(6) ln(Costs_2024)	(7) ln(Price_2024)	(8) ln(Price_2024)	(9) ln(Profit_2024)	(10) ln(Profit_2024)
Treated	0.150 (0.119)		0.164 (0.108)		-0.018 (0.107)		-0.027 (0.039)		0.261 (0.159)	
High HH Empowerment x Treated	-0.252* (0.140)		-0.244** (0.112)		-0.139 (0.136)		0.043 (0.051)		-0.425** (0.192)	
Treatment 1		0.093 (0.128)		0.101 (0.113)		-0.134 (0.110)		-0.014 (0.045)		0.183 (0.168)
Treatment 2		0.210 (0.152)		0.232* (0.136)		0.110 (0.127)		-0.042 (0.049)		0.346* (0.180)
High HH Empowerment x T1		-0.150 (0.148)		-0.160 (0.121)		-0.068 (0.143)		0.057 (0.045)		-0.329 (0.238)
High HH Empowerment x T2		-0.354** (0.179)		-0.331** (0.153)		-0.221 (0.172)		0.031 (0.075)		-0.526** (0.234)
Observations	977	977	977	977	977	977	977	977	977	977

Note: ln(Sales), ln(Quantity), ln(Costs), ln(Price) and ln(Profit) are the natural logs of total sales value, quantity sold, cultivation/harvest costs, unit price received, and profit, each measured per hectare of total cultivated area. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for baseline household empowerment, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.10.

Table 50: Heterogeneous Treatment Effects: Percentage of Sales by Buyer Type, by household empowerment

	CA 2024 (1)	(2)	Informal Agent 2024 (3)	(4)	Other 2024 (5)	(6)
Treated	0.002 (0.072)		0.021 (0.068)		-0.025 (0.018)	
Treatment 1		-0.064 (0.067)		0.073 (0.066)		-0.008 (0.020)
Treatment 2		0.075 (0.094)		-0.035 (0.088)		-0.043** (0.019)
High HH Empowerment x Treated	-0.047 (0.058)		0.031 (0.059)		0.015 (0.028)	
High HH Empowerment x T1		0.004 (0.061)		-0.019 (0.062)		0.013 (0.032)
High HH Empowerment x T2		-0.104 (0.066)		0.084 (0.069)		0.019 (0.032)
Observations	977	977	977	977	977	977

Note: Outcomes are the share of the farmer's total sales value sold to the Commission Agent (CA), to an informal agent, and to other buyer types. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for baseline household empowerment, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 51: Heterogeneous Treatment Effects: Flexible Arrangements, New Agent Consideration and Buyer Change, by household empowerment

	Flexible 2024		New Agent 2024		Change Buyer 2024	
	(1)	(2)	(3)	(4)	(5)	(6)
Treated	-0.011 (0.060)		0.165 (0.417)		-0.015 (0.079)	
Treatment 1		-0.013 (0.062)		0.418 (0.516)		0.062 (0.075)
Treatment 2		-0.009 (0.080)		-0.121 (0.481)		-0.105 (0.106)
High HH Empowerment x Treated	0.020 (0.055)		-0.675 (0.411)		0.006 (0.053)	
High HH Empowerment x T1		0.056 (0.055)		-0.669 (0.545)		0.046 (0.059)
High HH Empowerment x T2		-0.014 (0.073)		-0.644 (0.425)		-0.019 (0.060)
Observations	977	977	977	977	977	977

Note: Change Buyer is a binary indicator equal to one if the farmer sold to a different buyer/agent than at baseline. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for baseline household empowerment, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 52: Heterogeneous Treatment Effects: Harvest Costs, by household empowerment

	(1) ln(Harvest Cost Farmer '24)	(2) ln(Harvest Cost Farmer '24)	(3) ln(Harvest Cost Buyer '24)	(4) ln(Harvest Cost Buyer '24)	(5) Harvest Cost Share '24	(6) Harvest Cost Share '24
Treated	-0.284 (0.197)		0.476 (0.495)		-0.041 (0.027)	
High HH Empowerment x Treated	0.128 (0.207)		-0.566 (0.378)		0.064** (0.027)	
Treatment 1		-0.421** (0.213)		1.353** (0.595)		-0.096*** (0.033)
Treatment 2		-0.127 (0.262)		-0.503 (0.527)		0.020 (0.028)
High HH Empowerment x T1		0.066 (0.250)		-0.859 (0.544)		0.082** (0.039)
High HH Empowerment x T2		0.166 (0.220)		-0.164 (0.357)		0.039* (0.023)
Observations	977	977	977	977	977	977

Note: Outcomes are ln(Harvest Cost, Farmer), the log of harvest-time costs borne by the farmer, and Harvest Cost Farmer's Share, the farmer-paid share of total harvest costs (farmer plus buyer). Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for baseline household empowerment, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.10.

Table 53: Heterogeneous Treatment Effects: Additional Services Provided by Buyer, by household empowerment

	Additional Services 2024	
	(1)	(2)
Treated	0.053 (0.046)	
Treatment 1		0.156*** (0.043)
Treatment 2		-0.061 (0.049)
High HH Empowerment x Treated	-0.076* (0.040)	
High HH Empowerment x T1		-0.134*** (0.045)
High HH Empowerment x T2		-0.008 (0.039)
Observations	977	977

Note: Additional Services is a binary indicator equal to one if the buyer provided additional services (e.g. financial assistance or inputs) at cultivation. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for baseline household empowerment, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 54: Heterogeneous Treatment Effects: Agents Considered and Buyer Search Time, by household empowerment

	Agents Considered 2024		Buyer Search Time (Days) 2024	
	(1)	(2)	(3)	(4)
Treated	-0.102 (0.233)		-2.124 (1.891)	
Treatment 1		0.048 (0.263)		-1.046 (1.959)
Treatment 2		-0.268 (0.273)		-3.320 (2.172)
High HH Empowerment x Treated	-0.056 (0.193)		1.807 (2.139)	
High HH Empowerment x T1		-0.128 (0.215)		0.989 (2.211)
High HH Empowerment x T2		0.032 (0.203)		2.736 (2.373)
Observations	977	977	975	975

Note: Outcomes are the number of buyers/agents the farmer considered before selling, and the number of days taken to find a buyer and complete the sale. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for baseline household empowerment, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 55: Heterogeneous Treatment Effects: Sowing Choice, New Agent Consideration and Credit Constraint, by household empowerment

	Plant Potato (Expectation 2025)		New Agent (Expectation 2025)		Credit Constraint (Expectation 2025)	
	(1)	(2)	(3)	(4)	(5)	(6)
Treated	-0.007 (0.040)		0.017 (0.057)		0.078** (0.034)	
Treatment 1		-0.020 (0.042)		-0.008 (0.072)		0.110*** (0.042)
Treatment 2		0.010 (0.048)		0.048 (0.059)		0.042 (0.036)
High HH Empowerment x Treated	-0.037 (0.042)		-0.034 (0.064)		-0.001 (0.039)	
High HH Empowerment x T1		-0.074 (0.051)		-0.085 (0.074)		0.009 (0.056)
High HH Empowerment x T2		-0.006 (0.052)		0.010 (0.077)		-0.007 (0.041)
Observations	977	977	976	976	977	977

Note: Outcomes are Plant Potato (Expectation 2025) (intention to cultivate potatoes again in 2025), New Agent (Expectation 2025) (intention to consider a new Commission Agent), and Credit Constraint (Expectation 2025) (self-reported credit constraint), all reported at the 2024 endline, not realised 2025 outcomes. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for baseline household empowerment, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

3.6 Household Asset Index

Table 56: Heterogeneous Treatment Effects: Sales, Quantities, Prices and Profits, by household assets

	(1) ln(Sales_2024)	(2) ln(Sales_2024)	(3) ln(Quantity_2024)	(4) ln(Quantity_2024)	(5) ln(Costs_2024)	(6) ln(Costs_2024)	(7) ln(Price_2024)	(8) ln(Price_2024)	(9) ln(Profit_2024)	(10) ln(Profit_2024)
Treated	-0.100 (0.145)		-0.044 (0.108)		-0.209 (0.129)		-0.004 (0.041)		-0.132 (0.114)	
High Assets x Treated	0.227 (0.170)		0.157 (0.128)		0.209 (0.130)		-0.005 (0.070)		0.316 (0.198)	
Treatment 1		-0.089 (0.173)		-0.050 (0.125)		-0.283** (0.144)		0.010 (0.057)		-0.192 (0.143)
Treatment 2		-0.110 (0.148)		-0.037 (0.112)		-0.128 (0.136)		-0.020 (0.043)		-0.068 (0.143)
High Assets x T1		0.202 (0.200)		0.135 (0.145)		0.203 (0.155)		0.008 (0.080)		0.389* (0.222)
High Assets x T2		0.252 (0.182)		0.179 (0.152)		0.209 (0.141)		-0.016 (0.075)		0.240 (0.218)
Observations	977	977	977	977	977	977	977	977	977	977

Note: ln(Sales), ln(Quantity), ln(Costs), ln(Price) and ln(Profit) are the natural logs of total sales value, quantity sold, cultivation/harvest costs, unit price received, and profit, each measured per hectare of total cultivated area. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for the baseline household asset index, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.10.

Table 57: Heterogeneous Treatment Effects: Percentage of Sales by Buyer Type, by household assets

	CA 2024		Informal Agent 2024		Other 2024	
	(1)	(2)	(3)	(4)	(5)	(6)
Treated	0.027 (0.072)		-0.029 (0.069)		0.003 (0.015)	
Treatment 1		-0.026 (0.071)		0.010 (0.071)		0.018 (0.019)
Treatment 2		0.083 (0.093)		-0.071 (0.089)		-0.013 (0.018)
High Assets x Treated	-0.093* (0.052)		0.123** (0.049)		-0.034 (0.026)	
High Assets x T1		-0.067 (0.062)		0.096 (0.062)		-0.033 (0.031)
High Assets x T2		-0.122** (0.054)		0.152*** (0.051)		-0.034 (0.026)
Observations	977	977	977	977	977	977

Note: Outcomes are the share of the farmer's total sales value sold to the Commission Agent (CA), to an informal agent, and to other buyer types. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for the baseline household asset index, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.10.

Table 58: Heterogeneous Treatment Effects: Flexible Arrangements, New Agent Consideration and Buyer Change, by household assets

	Flexible 2024		New Agent 2024		Change Buyer 2024	
	(1)	(2)	(3)	(4)	(5)	(6)
Treated	-0.010 (0.060)		0.100 (0.460)		-0.038 (0.071)	
Treatment 1		0.018 (0.065)		0.452 (0.565)		0.050 (0.069)
Treatment 2		-0.040 (0.069)		-0.278 (0.517)		-0.135 (0.089)
High Assets x Treated	0.020 (0.061)		-0.548 (0.417)		0.048 (0.064)	
High Assets x T1		-0.004 (0.064)		-0.739 (0.568)		0.069 (0.075)
High Assets x T2		0.044 (0.069)		-0.336 (0.401)		0.034 (0.069)
Observations	977	977	977	977	977	977

Note: Change Buyer is a binary indicator equal to one if the farmer sold to a different buyer/agent than at baseline. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for the baseline household asset index, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.10.

Table 59: Heterogeneous Treatment Effects: Harvest Costs, by household assets

	(1)	(2)	(3)	(4)	(5)	(6)
	ln(Harvest Cost Farmer '24)	ln(Harvest Cost Farmer '24)	ln(Harvest Cost Buyer '24)	ln(Harvest Cost Buyer '24)	Harvest Cost Share '24	Harvest Cost Share '24
Treated	-0.208 (0.195)		0.382 (0.455)		-0.024 (0.025)	
High Assets x Treated	0.027 (0.275)		-0.382 (0.443)		0.028 (0.025)	
Treatment 1		-0.309 (0.225)		1.170** (0.521)		-0.063** (0.031)
Treatment 2		-0.092 (0.272)		-0.475 (0.451)		0.020 (0.027)
High Assets x T1		-0.112 (0.321)		-0.510 (0.588)		0.016 (0.031)
High Assets x T2		0.154 (0.343)		-0.199 (0.411)		0.036 (0.028)
Observations	977	977	977	977	977	977

Note: Outcomes are ln(Harvest Cost, Farmer), the log of harvest-time costs borne by the farmer, and Harvest Cost Farmer's Share, the farmer-paid share of total harvest costs (farmer plus buyer). Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for the baseline household asset index, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.10.

Table 60: Heterogeneous Treatment Effects: Additional Services Provided by Buyer, by household assets

	Additional Services 2024	
	(1)	(2)
Treated	0.033 (0.045)	
Treatment 1		0.118** (0.052)
Treatment 2		-0.058 (0.041)
High Assets x Treated	-0.037 (0.047)	
High Assets x T1		-0.060 (0.067)
High Assets x T2		-0.008 (0.039)
Observations	977	977

Note: Additional Services is a binary indicator equal to one if the buyer provided additional services (e.g. financial assistance or inputs) at cultivation. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for the baseline household asset index, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 61: Heterogeneous Treatment Effects: Agents Considered and Buyer Search Time, by household assets

	Agents Considered 2024		Buyer Search Time (Days) 2024	
	(1)	(2)	(3)	(4)
Treated	-0.183 (0.198)		-1.568 (2.390)	
Treatment 1		-0.141 (0.213)		-0.027 (2.389)
Treatment 2		-0.234 (0.255)		-3.181 (2.446)
High Assets x Treated	0.112 (0.145)		0.574 (3.019)	
High Assets x T1		0.254 (0.167)		-1.230 (3.055)
High Assets x T2		-0.024 (0.175)		2.439 (3.032)
Observations	977	977	975	975

Note: Outcomes are the number of buyers/agents the farmer considered before selling, and the number of days taken to find a buyer and complete the sale. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for the baseline household asset index, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 62: Heterogeneous Treatment Effects: Sowing Choice, New Agent Consideration and Credit Constraint, by household assets

	Plant Potato (Expectation 2025)		New Agent (Expectation 2025)		Credit Constraint (Expectation 2025)	
	(1)	(2)	(3)	(4)	(5)	(6)
Treated	-0.033 (0.049)		0.053 (0.064)		0.083** (0.034)	
Treatment 1		-0.057 (0.062)		-0.012 (0.074)		0.116*** (0.035)
Treatment 2		-0.007 (0.049)		0.123** (0.062)		0.046 (0.040)
High Assets x Treated	0.021 (0.040)		-0.100 (0.066)		-0.007 (0.031)	
High Assets x T1		0.003 (0.051)		-0.073 (0.075)		-0.001 (0.042)
High Assets x T2		0.037 (0.046)		-0.130* (0.074)		-0.011 (0.033)
Observations	977	977	976	976	977	977

Note: Outcomes are Plant Potato (Expectation 2025) (intention to cultivate potatoes again in 2025), New Agent (Expectation 2025) (intention to consider a new Commission Agent), and Credit Constraint (Expectation 2025) (self-reported credit constraint), all reported at the 2024 endline, not realised 2025 outcomes. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for the baseline household asset index, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.10.

3.7 Smartphone Access

Table 63: Heterogeneous Treatment Effects: Sales, Quantities, Prices and Profits, by smartphone access

	(1) ln(Sales_2024)	(2) ln(Sales_2024)	(3) ln(Quantity_2024)	(4) ln(Quantity_2024)	(5) ln(Costs_2024)	(6) ln(Costs_2024)	(7) ln(Price_2024)	(8) ln(Price_2024)	(9) ln(Profit_2024)	(10) ln(Profit_2024)
Treated	-0.066 (0.210)		-0.098 (0.154)		-0.305* (0.160)		0.054 (0.054)		-0.117 (0.142)	
Smartphone Access x Treated	0.119 (0.209)		0.196 (0.139)		0.306** (0.143)		-0.087 (0.066)		0.226 (0.193)	
Treatment 1		-0.103 (0.235)		-0.140 (0.163)		-0.390** (0.176)		0.065 (0.063)		-0.233 (0.176)
Treatment 2		-0.027 (0.212)		-0.055 (0.160)		-0.213 (0.165)		0.041 (0.058)		0.004 (0.137)
Smartphone Access x T1		0.170 (0.229)		0.230 (0.145)		0.316** (0.152)		-0.074 (0.070)		0.354* (0.205)
Smartphone Access x T2		0.067 (0.247)		0.159 (0.175)		0.290* (0.175)		-0.098 (0.076)		0.092 (0.235)
Observations	977	977	977	977	977	977	977	977	977	977

Note: ln(Sales), ln(Quantity), ln(Costs), ln(Price) and ln(Profit) are the natural logs of total sales value, quantity sold, cultivation/harvest costs, unit price received, and profit, each measured per hectare of total cultivated area. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for baseline smartphone access, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.10.

Table 64: Heterogeneous Treatment Effects: Percentage of Sales by Buyer Type, by smartphone access

	CA 2024 (1)	(2)	Informal Agent 2024 (3)	(4)	Other 2024 (5)	(6)
Treated	0.041 (0.081)		-0.037 (0.079)		-0.010 (0.016)	
Treatment 1		0.020 (0.083)		-0.017 (0.081)		-0.008 (0.019)
Treatment 2		0.065 (0.112)		-0.060 (0.109)		-0.013 (0.021)
Smartphone Access x Treated	-0.088 (0.070)		0.104 (0.069)		-0.009 (0.025)	
Smartphone Access x T1		-0.115 (0.077)		0.111 (0.074)		0.009 (0.026)
Smartphone Access x T2		-0.064 (0.079)		0.098 (0.083)		-0.026 (0.030)
Observations	977	977	977	977	977	977

Note: Outcomes are the share of the farmer's total sales value sold to the Commission Agent (CA), to an informal agent, and to other buyer types. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for baseline smartphone access, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 65: Heterogeneous Treatment Effects: Flexible Arrangements, New Agent Consideration and Buyer Change, by smartphone access

	Flexible 2024 (1)	(2)	New Agent 2024 (3)	(4)	Change Buyer 2024 (5)	(6)
Treated	0.042 (0.059)		-0.174 (0.513)		-0.088 (0.069)	
Treatment 1		0.069 (0.060)		0.229 (0.682)		0.029 (0.074)
Treatment 2		0.013 (0.073)		-0.609 (0.545)		-0.217*** (0.077)
Smartphone Access x Treated	-0.060 (0.049)		0.005 (0.392)		0.108 (0.077)	
Smartphone Access x T1		-0.074 (0.052)		-0.194 (0.600)		0.081 (0.083)
Smartphone Access x T2		-0.043 (0.060)		0.231 (0.351)		0.143 (0.091)
Observations	977	977	977	977	977	977

Note: Change Buyer is a binary indicator equal to one if the farmer sold to a different buyer/agent than at baseline. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for baseline smartphone access, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 66: Heterogeneous Treatment Effects: Harvest Costs, by smartphone access

	(1) ln(Harvest Cost Farmer '24)	(2) ln(Harvest Cost Farmer '24)	(3) ln(Harvest Cost Buyer '24)	(4) ln(Harvest Cost Buyer '24)	(5) Harvest Cost Share '24	(6) Harvest Cost Share '24
Treated	-0.464 (0.284)		0.815 (0.516)		-0.058** (0.027)	
Smartphone Access x Treated	0.332 (0.304)		-0.892* (0.483)		0.068** (0.030)	
Treatment 1		-0.380 (0.294)		1.529** (0.632)		-0.085*** (0.033)
Treatment 2		-0.537 (0.359)		0.023 (0.658)		-0.027 (0.039)
Smartphone Access x T1		-0.020 (0.337)		-0.877 (0.601)		0.043 (0.036)
Smartphone Access x T2		0.683** (0.294)		-0.853 (0.529)		0.092*** (0.032)
Observations	977	977	977	977	977	977

Note: Outcomes are ln(Harvest Cost, Farmer), the log of harvest-time costs borne by the farmer, and Harvest Cost Farmer's Share, the farmer-paid share of total harvest costs (farmer plus buyer). Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for baseline smartphone access, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.10.

Table 67: Heterogeneous Treatment Effects: Additional Services Provided by Buyer, by smartphone access

	Additional Services 2024	
	(1)	(2)
Treated	0.034 (0.047)	
Treatment 1		0.107* (0.055)
Treatment 2		-0.046 (0.059)
Smartphone Access x Treated	-0.027 (0.043)	
Smartphone Access x T1		-0.027 (0.056)
Smartphone Access x T2		-0.023 (0.049)
Observations	977	977

Note: Additional Services is a binary indicator equal to one if the buyer provided additional services (e.g. financial assistance or inputs) at cultivation. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for baseline smartphone access, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 68: Heterogeneous Treatment Effects: Agents Considered and Buyer Search Time, by smartphone access

	Agents Considered 2024		Buyer Search Time (Days) 2024	
	(1)	(2)	(3)	(4)
Treated	-0.302 (0.191)		-2.041 (2.338)	
Treatment 1		-0.295 (0.219)		-1.153 (2.446)
Treatment 2		-0.319 (0.243)		-3.005 (2.531)
Smartphone Access x Treated	0.245 (0.183)		1.107 (2.504)	
Smartphone Access x T1		0.392** (0.200)		0.774 (2.671)
Smartphone Access x T2		0.101 (0.191)		1.499 (2.697)
Observations	977	977	975	975

Note: Outcomes are the number of buyers/agents the farmer considered before selling, and the number of days taken to find a buyer and complete the sale. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for baseline smartphone access, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 69: Heterogeneous Treatment Effects: Sowing Choice, New Agent Consideration and Credit Constraint, by smartphone access

	Plant Potato (Expectation 2025)		New Agent (Expectation 2025)		Credit Constraint (Expectation 2025)	
	(1)	(2)	(3)	(4)	(5)	(6)
Treated	-0.003 (0.051)		0.115 (0.077)		0.058 (0.048)	
Treatment 1		-0.045 (0.059)		0.080 (0.101)		0.068 (0.055)
Treatment 2		0.043 (0.056)		0.154* (0.079)		0.045 (0.052)
Smartphone Access x Treated	-0.032 (0.049)		-0.162* (0.084)		0.028 (0.041)	
Smartphone Access x T1		-0.019 (0.052)		-0.186* (0.108)		0.065 (0.053)
Smartphone Access x T2		-0.048 (0.061)		-0.141* (0.086)		-0.009 (0.037)
Observations	977	977	976	976	977	977

Note: Outcomes are Plant Potato (Expectation 2025) (intention to cultivate potatoes again in 2025), New Agent (Expectation 2025) (intention to consider a new Commission Agent), and Credit Constraint (Expectation 2025) (self-reported credit constraint), all reported at the 2024 endline, not realised 2025 outcomes. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for baseline smartphone access, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.10.

3.8 Agricultural Income

Table 70: Heterogeneous Treatment Effects: Sales, Quantities, Prices and Profits, by agricultural income

	(1) ln(Sales_2024)	(2) ln(Sales_2024)	(3) ln(Quantity_2024)	(4) ln(Quantity_2024)	(5) ln(Costs_2024)	(6) ln(Costs_2024)	(7) ln(Price_2024)	(8) ln(Price_2024)	(9) ln(Profit_2024)	(10) ln(Profit_2024)
Treated	-0.062 (0.135)		-0.093 (0.106)		-0.202** (0.090)		0.032 (0.047)		-0.003 (0.082)	
High Agri Income x Treated	0.138 (0.159)		0.229* (0.122)		0.172 (0.120)		-0.067 (0.057)		0.023 (0.211)	
Treatment 1		-0.078 (0.157)		-0.136 (0.120)		-0.295*** (0.111)		0.060 (0.057)		-0.054 (0.107)
Treatment 2		-0.050 (0.144)		-0.054 (0.111)		-0.114 (0.091)		0.005 (0.049)		0.044 (0.083)
High Agri Income x T1		0.182 (0.156)		0.284** (0.132)		0.202 (0.131)		-0.070 (0.060)		0.093 (0.219)
High Agri Income x T2		0.092 (0.224)		0.176 (0.144)		0.158 (0.138)		-0.069 (0.080)		-0.045 (0.310)
Observations	967	967	967	967	967	967	967	967	967	967

Note: ln(Sales), ln(Quantity), ln(Costs), ln(Price) and ln(Profit) are the natural logs of total sales value, quantity sold, cultivation/harvest costs, unit price received, and profit, each measured per hectare of total cultivated area. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for baseline agricultural income, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.10.

Table 71: Heterogeneous Treatment Effects: Percentage of Sales by Buyer Type, by agricultural income

	CA 2024		Informal Agent 2024		Other 2024	
	(1)	(2)	(3)	(4)	(5)	(6)
Treated	-0.037 (0.066)		0.011 (0.067)		0.024 (0.015)	
Treatment 1		-0.101* (0.059)		0.058 (0.061)		0.042* (0.023)
Treatment 2		0.023 (0.094)		-0.033 (0.095)		0.007 (0.014)
High Agri Income x Treated	0.039 (0.058)		0.040 (0.064)		-0.079*** (0.027)	
High Agri Income x T1		0.082 (0.063)		0.007 (0.071)		-0.088*** (0.031)
High Agri Income x T2		0.004 (0.065)		0.068 (0.072)		-0.073** (0.031)
Observations	967	967	967	967	967	967

Note: Outcomes are the share of the farmer's total sales value sold to the Commission Agent (CA), to an informal agent, and to other buyer types. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for baseline agricultural income, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.10.

Table 72: Heterogeneous Treatment Effects: Flexible Arrangements, New Agent Consideration and Buyer Change, by agricultural income

	Flexible 2024 (1)	(2)	New Agent 2024 (3)	(4)	Change Buyer 2024 (5)	(6)
Treated	-0.022 (0.054)		0.047 (0.390)		-0.002 (0.074)	
Treatment 1		0.015 (0.061)		0.373 (0.506)		0.115 (0.071)
Treatment 2		-0.055 (0.066)		-0.265 (0.434)		-0.114 (0.090)
High Agri Income x Treated	0.048 (0.048)		-0.445 (0.394)		-0.026 (0.067)	
High Agri Income x T1		0.006 (0.056)		-0.560 (0.541)		-0.059 (0.075)
High Agri Income x T2		0.087* (0.052)		-0.380 (0.390)		-0.012 (0.076)
Observations	967	967	967	967	967	967

Note: Change Buyer is a binary indicator equal to one if the farmer sold to a different buyer/agent than at baseline. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for baseline agricultural income, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 73: Heterogeneous Treatment Effects: Harvest Costs, by agricultural income

	(1)	(2)	(3)	(4)	(5)	(6)
	ln(Harvest Cost Farmer '24)	ln(Harvest Cost Farmer '24)	ln(Harvest Cost Buyer '24)	ln(Harvest Cost Buyer '24)	Harvest Cost Share '24	Harvest Cost Share '24
Treated	-0.284 (0.261)		0.110 (0.556)		-0.006 (0.036)	
High Agri Income x Treated	0.242 (0.215)		0.103 (0.542)		-0.001 (0.033)	
Treatment 1		-0.553* (0.289)		0.710 (0.684)		-0.057 (0.038)
Treatment 2		-0.027 (0.304)		-0.487 (0.600)		0.043 (0.040)
High Agri Income x T1		0.377 (0.272)		0.409 (0.746)		0.005 (0.037)
High Agri Income x T2		0.143 (0.209)		-0.359 (0.504)		0.001 (0.035)
Observations	967	967	967	967	967	967

Note: Outcomes are ln(Harvest Cost, Farmer), the log of harvest-time costs borne by the farmer, and Harvest Cost Farmer's Share, the farmer-paid share of total harvest costs (farmer plus buyer). Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for baseline agricultural income, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.10.

Table 74: Heterogeneous Treatment Effects: Additional Services Provided by Buyer, by agricultural income

	Additional Services 2024	
	(1)	(2)
Treated	0.033 (0.048)	
Treatment 1		0.113** (0.053)
Treatment 2		-0.045 (0.052)
High Agri Income x Treated	-0.033 (0.053)	
High Agri Income x T1		-0.043 (0.066)
High Agri Income x T2		-0.037 (0.052)
Observations	967	967

Note: Additional Services is a binary indicator equal to one if the buyer provided additional services (e.g. financial assistance or inputs) at cultivation. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for baseline agricultural income, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 75: Heterogeneous Treatment Effects: Agents Considered and Buyer Search Time, by agricultural income

	Agents Considered 2024		Buyer Search Time (Days) 2024	
	(1)	(2)	(3)	(4)
Treated	-0.170 (0.184)		0.014 (1.758)	
Treatment 1		-0.044 (0.205)		1.106 (1.850)
Treatment 2		-0.291 (0.243)		-1.007 (1.910)
High Agri Income x Treated	0.066 (0.205)		-2.119 (1.979)	
High Agri Income x T1		0.042 (0.225)		-3.022 (2.135)
High Agri Income x T2		0.069 (0.223)		-1.320 (2.129)
Observations	967	967	966	966

Note: Outcomes are the number of buyers/agents the farmer considered before selling, and the number of days taken to find a buyer and complete the sale. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for baseline agricultural income, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 76: Heterogeneous Treatment Effects: Sowing Choice, New Agent Consideration and Credit Constraint, by agricultural income

	Plant Potato (Expectation 2025)		New Agent (Expectation 2025)		Credit Constraint (Expectation 2025)	
	(1)	(2)	(3)	(4)	(5)	(6)
Treated	-0.034 (0.047)		0.007 (0.057)		0.106*** (0.031)	
Treatment 1		-0.055 (0.056)		-0.078 (0.066)		0.160*** (0.037)
Treatment 2		-0.013 (0.047)		0.086 (0.057)		0.055* (0.031)
High Agri Income x Treated	0.023 (0.050)		-0.001 (0.069)		-0.052* (0.027)	
High Agri Income x T1		-0.002 (0.059)		0.063 (0.089)		-0.086** (0.035)
High Agri Income x T2		0.054 (0.051)		-0.056 (0.077)		-0.024 (0.042)
Observations	967	967	966	966	967	967

Note: Outcomes are Plant Potato (Expectation 2025) (intention to cultivate potatoes again in 2025), New Agent (Expectation 2025) (intention to consider a new Commission Agent), and Credit Constraint (Expectation 2025) (self-reported credit constraint), all reported at the 2024 endline, not realised 2025 outcomes. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for baseline agricultural income, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

3.9 Non-Agricultural Income

Table 77: Heterogeneous Treatment Effects: Sales, Quantities, Prices and Profits, by non-agricultural income

	(1) ln(Sales_2024)	(2) ln(Sales_2024)	(3) ln(Quantity_2024)	(4) ln(Quantity_2024)	(5) ln(Costs_2024)	(6) ln(Costs_2024)	(7) ln(Price_2024)	(8) ln(Price_2024)	(9) ln(Profit_2024)	(10) ln(Profit_2024)
Treated	-0.018 (0.142)		-0.014 (0.128)		-0.024 (0.129)		0.010 (0.049)		-0.071 (0.170)	
High NonAgri Income x Treated	0.084 (0.154)		0.115 (0.130)		-0.177 (0.159)		-0.035 (0.067)		0.280 (0.230)	
Treatment 1		-0.026 (0.147)		-0.059 (0.124)		-0.110 (0.119)		0.043 (0.057)		-0.201 (0.197)
Treatment 2		-0.011 (0.162)		0.028 (0.152)		0.053 (0.154)		-0.019 (0.053)		0.049 (0.186)
High NonAgri Income x T1		0.116 (0.153)		0.186 (0.122)		-0.158 (0.158)		-0.050 (0.068)		0.521** (0.253)
High NonAgri Income x T2		0.044 (0.196)		0.039 (0.168)		-0.158 (0.203)		-0.032 (0.085)		0.014 (0.281)
Observations	967	967	967	967	967	967	967	967	967	967

Note: ln(Sales), ln(Quantity), ln(Costs), ln(Price) and ln(Profit) are the natural logs of total sales value, quantity sold, cultivation/harvest costs, unit price received, and profit, each measured per hectare of total cultivated area. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for baseline non-agricultural income, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.10.

Table 78: Heterogeneous Treatment Effects: Percentage of Sales by Buyer Type, by non-agricultural income

	CA 2024 (1)	(2)	Informal Agent 2024 (3)	(4)	Other 2024 (5)	(6)
Treated	-0.004 (0.081)		0.041 (0.075)		-0.038** (0.019)	
Treatment 1		-0.052 (0.069)		0.086 (0.066)		-0.032 (0.022)
Treatment 2		0.040 (0.110)		0.001 (0.104)		-0.044** (0.021)
High NonAgri Income x Treated	-0.050 (0.090)		-0.003 (0.084)		0.050** (0.021)	
High NonAgri Income x T1		-0.021 (0.071)		-0.047 (0.067)		0.063** (0.026)
High NonAgri Income x T2		-0.063 (0.121)		0.034 (0.117)		0.029 (0.020)
Observations	967	967	967	967	967	967

Note: Outcomes are the share of the farmer's total sales value sold to the Commission Agent (CA), to an informal agent, and to other buyer types. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for baseline non-agricultural income, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 79: Heterogeneous Treatment Effects: Flexible Arrangements, New Agent Consideration and Buyer Change, by non-agricultural income

	Flexible 2024 (1)	(2)	New Agent 2024 (3)	(4)	Change Buyer 2024 (5)	(6)
Treated	-0.005 (0.068)		-0.287 (0.476)		-0.002 (0.088)	
Treatment 1		0.005 (0.072)		0.186 (0.515)		0.099 (0.075)
Treatment 2		-0.012 (0.079)		-0.716 (0.541)		-0.093 (0.117)
High NonAgri Income x Treated	0.023 (0.066)		0.301 (0.548)		-0.025 (0.090)	
High NonAgri Income x T1		0.035 (0.070)		-0.245 (0.572)		-0.034 (0.074)
High NonAgri Income x T2		-0.001 (0.076)		0.809 (0.620)		-0.066 (0.113)
Observations	967	967	967	967	967	967

Note: Change Buyer is a binary indicator equal to one if the farmer sold to a different buyer/agent than at baseline. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for baseline non-agricultural income, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.10.

Table 80: Heterogeneous Treatment Effects: Harvest Costs, by non-agricultural income

	(1)	(2)	(3)	(4)	(5)	(6)
	ln(Harvest Cost Farmer '24)	ln(Harvest Cost Farmer '24)	ln(Harvest Cost Buyer '24)	ln(Harvest Cost Buyer '24)	Harvest Cost Share '24	Harvest Cost Share '24
Treated	-0.212 (0.282)		0.036 (0.513)		-0.005 (0.029)	
High NonAgri Income x Treated	-0.099 (0.311)		0.340 (0.699)		-0.009 (0.037)	
Treatment 1		-0.370 (0.277)		0.981 (0.603)		-0.060* (0.032)
Treatment 2		-0.072 (0.345)		-0.808 (0.524)		0.045 (0.030)
High NonAgri Income x T1		-0.111 (0.337)		-0.146 (0.815)		0.012 (0.042)
High NonAgri Income x T2		0.000 (0.341)		0.509 (0.610)		-0.010 (0.034)
Observations	967	967	967	967	967	967

Note: Outcomes are ln(Harvest Cost, Farmer), the log of harvest-time costs borne by the farmer, and Harvest Cost Farmer's Share, the farmer-paid share of total harvest costs (farmer plus buyer). Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for baseline non-agricultural income, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.10.

Table 81: Heterogeneous Treatment Effects: Additional Services Provided by Buyer, by non-agricultural income

	Additional Services 2024	
	(1)	(2)
Treated	0.004 (0.046)	
Treatment 1		0.092* (0.049)
Treatment 2		-0.075 (0.048)
High NonAgri Income x Treated	0.033 (0.065)	
High NonAgri Income x T1		-0.002 (0.075)
High NonAgri Income x T2		0.035 (0.060)
Observations	967	967

Note: Additional Services is a binary indicator equal to one if the buyer provided additional services (e.g. financial assistance or inputs) at cultivation. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for baseline non-agricultural income, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 82: Heterogeneous Treatment Effects: Agents Considered and Buyer Search Time, by non-agricultural income

	Agents Considered 2024		Buyer Search Time (Days) 2024	
	(1)	(2)	(3)	(4)
Treated	-0.060 (0.232)		-1.468 (1.336)	
Treatment 1		0.095 (0.245)		-0.747 (1.348)
Treatment 2		-0.198 (0.287)		-2.121 (1.526)
High NonAgri Income x Treated	-0.176 (0.248)		1.020 (1.993)	
High NonAgri Income x T1		-0.281 (0.243)		0.581 (2.154)
High NonAgri Income x T2		-0.111 (0.292)		1.250 (2.394)
Observations	967	967	966	966

Note: Outcomes are the number of buyers/agents the farmer considered before selling, and the number of days taken to find a buyer and complete the sale. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for baseline non-agricultural income, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 83: Heterogeneous Treatment Effects: Sowing Choice, New Agent Consideration and Credit Constraint, by non-agricultural income

	Plant Potato (Expectation 2025)		New Agent (Expectation 2025)		Credit Constraint (Expectation 2025)	
	(1)	(2)	(3)	(4)	(5)	(6)
Treated	-0.037 (0.042)		-0.027 (0.051)		0.050* (0.029)	
Treatment 1		-0.055 (0.056)		-0.078 (0.070)		0.073** (0.036)
Treatment 2		-0.020 (0.042)		0.018 (0.051)		0.030 (0.032)
High NonAgri Income x Treated	0.025 (0.049)		0.064 (0.079)		0.075* (0.043)	
High NonAgri Income x T1		-0.006 (0.062)		0.067 (0.095)		0.101** (0.043)
High NonAgri Income x T2		0.078 (0.051)		0.088 (0.083)		0.026 (0.057)
Observations	967	967	966	966	967	967

Note: Outcomes are Plant Potato (Expectation 2025) (intention to cultivate potatoes again in 2025), New Agent (Expectation 2025) (intention to consider a new Commission Agent), and Credit Constraint (Expectation 2025) (self-reported credit constraint), all reported at the 2024 endline, not realised 2025 outcomes. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for baseline non-agricultural income, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

3.10 Trust

Table 84: Heterogeneous Treatment Effects: Sales, Quantities, Prices and Profits, by trust

	(1) ln(Sales_2024)	(2) ln(Sales_2024)	(3) ln(Quantity_2024)	(4) ln(Quantity_2024)	(5) ln(Costs_2024)	(6) ln(Costs_2024)	(7) ln(Price_2024)	(8) ln(Price_2024)	(9) ln(Profit_2024)	(10) ln(Profit_2024)
Treated	-0.092 (0.122)		0.057 (0.112)		-0.027 (0.123)		-0.076** (0.038)		-0.054 (0.169)	
High Trust x Treated	0.214 (0.140)		-0.032 (0.107)		-0.114 (0.118)		0.130** (0.061)		0.187 (0.191)	
Treatment 1		-0.114 (0.136)		0.006 (0.117)		-0.118 (0.128)		-0.061 (0.048)		-0.077 (0.165)
Treatment 2		-0.064 (0.146)		0.121 (0.139)		0.085 (0.146)		-0.094** (0.044)		-0.026 (0.250)
High Trust x T1		0.260* (0.140)		0.040 (0.111)		-0.092 (0.135)		0.142** (0.060)		0.184 (0.222)
High Trust x T2		0.165 (0.169)		-0.113 (0.141)		-0.164 (0.145)		0.124* (0.076)		0.182 (0.248)
Observations	977	977	977	977	977	977	977	977	977	977

Note: ln(Sales), ln(Quantity), ln(Costs), ln(Price) and ln(Profit) are the natural logs of total sales value, quantity sold, cultivation/harvest costs, unit price received, and profit, each measured per hectare of total cultivated area. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for baseline trust, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.10.

Table 85: Heterogeneous Treatment Effects: Percentage of Sales by Buyer Type, by trust

	CA 2024		Informal Agent 2024		Other 2024	
	(1)	(2)	(3)	(4)	(5)	(6)
Treated	-0.078 (0.068)		0.097 (0.062)		-0.020 (0.026)	
Treatment 1		-0.105 (0.069)		0.117* (0.067)		-0.013 (0.028)
Treatment 2		-0.046 (0.083)		0.074 (0.073)		-0.029 (0.032)
High Trust x Treated	0.108** (0.051)		-0.116** (0.051)		0.006 (0.028)	
High Trust x T1		0.083 (0.053)		-0.103* (0.055)		0.021 (0.030)
High Trust x T2		0.123* (0.063)		-0.123** (0.063)		-0.004 (0.034)
Observations	977	977	977	977	977	977

Note: Outcomes are the share of the farmer's total sales value sold to the Commission Agent (CA), to an informal agent, and to other buyer types. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for baseline trust, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.10.

Table 86: Heterogeneous Treatment Effects: Flexible Arrangements, New Agent Consideration and Buyer Change, by trust

	Flexible 2024		New Agent 2024		Change Buyer 2024	
	(1)	(2)	(3)	(4)	(5)	(6)
Treated	-0.008 (0.058)		0.125 (0.384)		-0.025 (0.077)	
Treatment 1		-0.003 (0.066)		0.515 (0.453)		0.028 (0.075)
Treatment 2		-0.014 (0.069)		-0.361 (0.392)		-0.089 (0.098)
High Trust x Treated	0.015 (0.050)		-0.566 (0.398)		0.022 (0.070)	
High Trust x T1		0.041 (0.057)		-0.858** (0.432)		0.117* (0.068)
High Trust x T2		-0.004 (0.057)		-0.174 (0.432)		-0.045 (0.081)
Observations	977	977	977	977	977	977

Note: Change Buyer is a binary indicator equal to one if the farmer sold to a different buyer/agent than at baseline. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for baseline trust, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 87: Heterogeneous Treatment Effects: Harvest Costs, by trust

	(1)	(2)	(3)	(4)	(5)	(6)
	ln(Harvest Cost Farmer '24)	ln(Harvest Cost Farmer '24)	ln(Harvest Cost Buyer '24)	ln(Harvest Cost Buyer '24)	Harvest Cost Share '24	Harvest Cost Share '24
Treated	-0.378* (0.220)		0.718 (0.445)		-0.045* (0.024)	
High Trust x Treated	0.280 (0.299)		-1.041** (0.446)		0.068** (0.030)	
Treatment 1		-0.491** (0.227)		1.206** (0.529)		-0.078*** (0.029)
Treatment 2		-0.240 (0.280)		0.116 (0.571)		-0.003 (0.031)
High Trust x T1		0.147 (0.298)		-0.569 (0.665)		0.042 (0.039)
High Trust x T2		0.361 (0.382)		-1.297*** (0.462)		0.079** (0.034)
Observations	977	977	977	977	977	977

Note: Outcomes are ln(Harvest Cost, Farmer), the log of harvest-time costs borne by the farmer, and Harvest Cost Farmer's Share, the farmer-paid share of total harvest costs (farmer plus buyer). Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for baseline trust, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 88: Heterogeneous Treatment Effects: Additional Services Provided by Buyer, by trust

	Additional Services 2024	
	(1)	(2)
Treated	0.066* (0.038)	
Treatment 1		0.113*** (0.041)
Treatment 2		0.009 (0.047)
High Trust x Treated	-0.101** (0.042)	
High Trust x T1		-0.047 (0.062)
High Trust x T2		-0.133*** (0.041)
Observations	977	977

Note: Additional Services is a binary indicator equal to one if the buyer provided additional services (e.g. financial assistance or inputs) at cultivation. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for baseline trust, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 89: Heterogeneous Treatment Effects: Agents Considered and Buyer Search Time, by trust

	Agents Considered 2024		Buyer Search Time (Days) 2024	
	(1)	(2)	(3)	(4)
Treated	0.071 (0.215)		-1.183 (1.941)	
Treatment 1		0.139 (0.237)		-0.044 (2.070)
Treatment 2		-0.012 (0.260)		-2.601 (1.894)
High Trust x Treated	-0.391** (0.194)		-0.158 (2.753)	
High Trust x T1		-0.312 (0.227)		-1.227 (3.047)
High Trust x T2		-0.438** (0.212)		1.175 (2.795)
Observations	977	977	975	975

Note: Outcomes are the number of buyers/agents the farmer considered before selling, and the number of days taken to find a buyer and complete the sale. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for baseline trust, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 90: Heterogeneous Treatment Effects: Sowing Choice, New Agent Consideration and Credit Constraint, by trust

	Plant Potato (Expectation 2025)		New Agent (Expectation 2025)		Credit Constraint (Expectation 2025)	
	(1)	(2)	(3)	(4)	(5)	(6)
Treated	-0.019 (0.047)		-0.019 (0.061)		0.069* (0.035)	
Treatment 1		-0.061 (0.055)		-0.071 (0.073)		0.100** (0.039)
Treatment 2		0.032 (0.047)		0.045 (0.062)		0.030 (0.038)
High Trust x Treated	-0.014 (0.042)		0.038 (0.050)		0.018 (0.035)	
High Trust x T1		0.003 (0.045)		0.043 (0.062)		0.031 (0.054)
High Trust x T2		-0.043 (0.058)		0.016 (0.058)		0.017 (0.030)
Observations	977	977	976	976	977	977

Note: Outcomes are Plant Potato (Expectation 2025) (intention to cultivate potatoes again in 2025), New Agent (Expectation 2025) (intention to consider a new Commission Agent), and Credit Constraint (Expectation 2025) (self-reported credit constraint), all reported at the 2024 endline, not realised 2025 outcomes. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for baseline trust, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.10.

3.11 Food Insecurity Index

Table 91: Heterogeneous Treatment Effects: Sales, Quantities, Prices and Profits, by food insecurity

	(1) ln(Sales_2024)	(2) ln(Sales_2024)	(3) ln(Quantity_2024)	(4) ln(Quantity_2024)	(5) ln(Costs_2024)	(6) ln(Costs_2024)	(7) ln(Price_2024)	(8) ln(Price_2024)	(9) ln(Profit_2024)	(10) ln(Profit_2024)
Treated	0.089 (0.136)		0.081 (0.103)		-0.030 (0.119)		0.011 (0.050)		0.193 (0.198)	
High Food Insecurity x Treated	-0.198* (0.117)		-0.112 (0.090)		-0.178 (0.127)		-0.050 (0.065)		-0.394** (0.185)	
Treatment 1		0.070 (0.143)		0.038 (0.114)		-0.131 (0.121)		0.055 (0.055)		0.225 (0.188)
Treatment 2		0.107 (0.156)		0.120 (0.109)		0.062 (0.131)		-0.029 (0.059)		0.164 (0.256)
High Food Insecurity x T1		-0.159 (0.130)		-0.060 (0.103)		-0.132 (0.138)		-0.097 (0.074)		-0.500** (0.224)
High Food Insecurity x T2		-0.237* (0.139)		-0.161 (0.103)		-0.201 (0.144)		-0.009 (0.076)		-0.280 (0.201)
Observations	977	977	977	977	977	977	977	977	977	977

Note: ln(Sales), ln(Quantity), ln(Costs), ln(Price) and ln(Profit) are the natural logs of total sales value, quantity sold, cultivation/harvest costs, unit price received, and profit, each measured per hectare of total cultivated area. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for the baseline food insecurity index, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.10.

Table 92: Heterogeneous Treatment Effects: Percentage of Sales by Buyer Type, by food insecurity

	CA 2024 (1)	(2)	Informal Agent 2024 (3)	(4)	Other 2024 (5)	(6)
Treated	-0.021 (0.064)		0.029 (0.063)		-0.010 (0.022)	
Treatment 1		-0.025 (0.062)		0.013 (0.063)		0.014 (0.025)
Treatment 2		-0.016 (0.092)		0.044 (0.090)		-0.031 (0.023)
High Food Insecurity x Treated	-0.014 (0.070)		0.022 (0.073)		-0.007 (0.036)	
High Food Insecurity x T1		-0.085 (0.067)		0.107 (0.071)		-0.024 (0.037)
High Food Insecurity x T2		0.068 (0.084)		-0.072 (0.085)		0.004 (0.037)
Observations	977	977	977	977	977	977

Note: Outcomes are the share of the farmer's total sales value sold to the Commission Agent (CA), to an informal agent, and to other buyer types. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for the baseline food insecurity index, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 93: Heterogeneous Treatment Effects: Flexible Arrangements, New Agent Consideration and Buyer Change, by food insecurity

	Flexible 2024 (1)	(2)	New Agent 2024 (3)	(4)	Change Buyer 2024 (5)	(6)
Treated	0.021 (0.060)		-0.519 (0.370)		-0.040 (0.078)	
Treatment 1		0.033 (0.061)		-0.359 (0.423)		0.061 (0.078)
Treatment 2		0.010 (0.075)		-0.668 (0.426)		-0.134 (0.091)
High Food Insecurity x Treated	-0.058 (0.047)		0.901** (0.455)		0.073 (0.076)	
High Food Insecurity x T1		-0.048 (0.049)		1.076* (0.616)		0.064 (0.092)
High Food Insecurity x T2		-0.074 (0.059)		0.641 (0.412)		0.050 (0.084)
Observations	977	977	977	977	977	977

Note: Change Buyer is a binary indicator equal to one if the farmer sold to a different buyer/agent than at baseline. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for the baseline food insecurity index, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.10.

Table 94: Heterogeneous Treatment Effects: Harvest Costs, by food insecurity

	(1)	(2)	(3)	(4)	(5)	(6)
	ln(Harvest Cost Farmer '24)	ln(Harvest Cost Farmer '24)	ln(Harvest Cost Buyer '24)	ln(Harvest Cost Buyer '24)	Harvest Cost Share '24	Harvest Cost Share '24
Treated	-0.124 (0.232)		-0.309 (0.434)		0.020 (0.028)	
High Food Insecurity x Treated	-0.256 (0.258)		1.338*** (0.367)		-0.078*** (0.027)	
Treatment 1		-0.260 (0.250)		0.631 (0.515)		-0.031 (0.030)
Treatment 2		0.003 (0.329)		-1.176*** (0.417)		0.067** (0.032)
High Food Insecurity x T1		-0.312 (0.314)		0.874** (0.445)		-0.065** (0.028)
High Food Insecurity x T2		-0.147 (0.324)		1.585*** (0.349)		-0.077** (0.030)
Observations	977	977	977	977	977	977

Note: Outcomes are ln(Harvest Cost, Farmer), the log of harvest-time costs borne by the farmer, and Harvest Cost Farmer's Share, the farmer-paid share of total harvest costs (farmer plus buyer). Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for the baseline food insecurity index, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.10.

Table 95: Heterogeneous Treatment Effects: Additional Services Provided by Buyer, by food insecurity

	Additional Services 2024	
	(1)	(2)
Treated	-0.032 (0.037)	
Treatment 1		0.055 (0.046)
Treatment 2		-0.111*** (0.034)
High Food Insecurity x Treated	0.127*** (0.044)	
High Food Insecurity x T1		0.098* (0.053)
High Food Insecurity x T2		0.135*** (0.043)
Observations	977	977

Note: Additional Services is a binary indicator equal to one if the buyer provided additional services (e.g. financial assistance or inputs) at cultivation. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for the baseline food insecurity index, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 96: Heterogeneous Treatment Effects: Agents Considered and Buyer Search Time, by food insecurity

	Agents Considered 2024		Buyer Search Time (Days) 2024	
	(1)	(2)	(3)	(4)
Treated	-0.176 (0.213)		-2.231 (1.617)	
Treatment 1		-0.011 (0.238)		-2.495 (1.787)
Treatment 2		-0.328 (0.248)		-1.991 (1.797)
High Food Insecurity x Treated	0.154 (0.250)		3.203** (1.574)	
High Food Insecurity x T1		0.046 (0.264)		5.008** (1.997)
High Food Insecurity x T2		0.228 (0.292)		1.146 (1.491)
Observations	977	977	975	975

Note: Outcomes are the number of buyers/agents the farmer considered before selling, and the number of days taken to find a buyer and complete the sale. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for the baseline food insecurity index, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 97: Heterogeneous Treatment Effects: Sowing Choice, New Agent Consideration and Credit Constraint, by food insecurity

	Plant Potato (Expectation 2025)		New Agent (Expectation 2025)		Credit Constraint (Expectation 2025)	
	(1)	(2)	(3)	(4)	(5)	(6)
Treated	-0.014 (0.042)		0.015 (0.056)		0.068** (0.034)	
Treatment 1		-0.055 (0.052)		-0.037 (0.077)		0.104*** (0.036)
Treatment 2		0.024 (0.041)		0.064 (0.058)		0.034 (0.041)
High Food Insecurity x Treated	-0.023 (0.039)		-0.029 (0.089)		0.018 (0.037)	
High Food Insecurity x T1		-0.003 (0.050)		-0.024 (0.107)		0.017 (0.050)
High Food Insecurity x T2		-0.032 (0.039)		-0.016 (0.091)		0.007 (0.039)
Observations	977	977	976	976	977	977

Note: Outcomes are Plant Potato (Expectation 2025) (intention to cultivate potatoes again in 2025), New Agent (Expectation 2025) (intention to consider a new Commission Agent), and Credit Constraint (Expectation 2025) (self-reported credit constraint), all reported at the 2024 endline, not realised 2025 outcomes. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for the baseline food insecurity index, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

3.12 Buyer Provided the Best Price in 2023

Table 98: Heterogeneous Treatment Effects: Sales, Quantities, Prices and Profits, by whether the buyer offered the best price in 2023

	(1) ln(Sales_2024)	(2) ln(Sales_2024)	(3) ln(Quantity_2024)	(4) ln(Quantity_2024)	(5) ln(Costs_2024)	(6) ln(Costs_2024)	(7) ln(Price_2024)	(8) ln(Price_2024)	(9) ln(Profit_2024)	(10) ln(Profit_2024)
Treated	0.285 (0.228)		0.176 (0.214)		0.152 (0.217)		0.026 (0.077)		0.168 (0.172)	
Best Price 2023 x Treated	-0.363 (0.239)		-0.171 (0.222)		-0.325 (0.219)		-0.053 (0.077)		-0.139 (0.242)	
Treatment 1		0.142 (0.270)		0.029 (0.227)		-0.056 (0.239)		0.019 (0.103)		-0.026 (0.299)
Treatment 2		0.368 (0.235)		0.259 (0.224)		0.267 (0.224)		0.031 (0.080)		0.277* (0.167)
Best Price 2023 x T1		-0.160 (0.289)		0.010 (0.246)		-0.138 (0.247)		-0.018 (0.099)		0.099 (0.355)
Best Price 2023 x T2		-0.518* (0.269)		-0.292 (0.234)		-0.403* (0.229)		-0.094 (0.086)		-0.298 (0.305)
Observations	977	977	977	977	977	977	977	977	977	977

Note: ln(Sales), ln(Quantity), ln(Costs), ln(Price) and ln(Profit) are the natural logs of total sales value, quantity sold, cultivation/harvest costs, unit price received, and profit, each measured per hectare of total cultivated area. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for whether the buyer offered the best price in 2023, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.10.

Table 99: Heterogeneous Treatment Effects: Percentage of Sales by Buyer Type, by whether the buyer offered the best price in 2023

	CA 2024		Informal Agent 2024		Other 2024	
	(1)	(2)	(3)	(4)	(5)	(6)
Treated	-0.003 (0.108)		-0.021 (0.102)		0.014 (0.025)	
Treatment 1		-0.085 (0.091)		0.012 (0.096)		0.063 (0.043)
Treatment 2		0.040 (0.134)		-0.038 (0.129)		-0.013 (0.021)
Best Price 2023 x Treated	-0.024 (0.106)		0.080 (0.104)		-0.044 (0.033)	
Best Price 2023 x T1		0.033 (0.098)		0.066 (0.106)		-0.086* (0.046)
Best Price 2023 x T2		-0.031 (0.120)		0.071 (0.115)		-0.029 (0.035)
Observations	977	977	977	977	977	977

Note: Outcomes are the share of the farmer's total sales value sold to the Commission Agent (CA), to an informal agent, and to other buyer types. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for whether the buyer offered the best price in 2023, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 100: Heterogeneous Treatment Effects: Flexible Arrangements, New Agent Consideration and Buyer Change, by whether the buyer offered the best price in 2023

	Flexible 2024		New Agent 2024		Change Buyer 2024	
	(1)	(2)	(3)	(4)	(5)	(6)
Treated	0.074 (0.086)		-0.559 (0.568)		-0.047 (0.108)	
Treatment 1		0.100 (0.084)		-0.125 (0.550)		0.082 (0.110)
Treatment 2		0.061 (0.093)		-0.792 (0.624)		-0.115 (0.113)
Best Price 2023 x Treated	-0.103 (0.088)		0.550 (0.598)		0.053 (0.107)	
Best Price 2023 x T1		-0.108 (0.085)		0.295 (0.697)		0.009 (0.123)
Best Price 2023 x T2		-0.116 (0.099)		0.527 (0.568)		0.001 (0.111)
Observations	977	977	977	977	977	977

Note: Change Buyer is a binary indicator equal to one if the farmer sold to a different buyer/agent than at baseline. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for whether the buyer offered the best price in 2023, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 101: Heterogeneous Treatment Effects: Harvest Costs, by whether the buyer offered the best price in 2023

	(1) ln(Harvest Cost Farmer '24)	(2) ln(Harvest Cost Farmer '24)	(3) ln(Harvest Cost Buyer '24)	(4) ln(Harvest Cost Buyer '24)	(5) Harvest Cost Share '24	(6) Harvest Cost Share '24
Treated	-0.165 (0.365)		-1.129* (0.658)		0.046 (0.033)	
Best Price 2023 x Treated	-0.059 (0.374)		1.863*** (0.622)		-0.076** (0.033)	
Treatment 1		-0.676* (0.403)		-0.346 (0.750)		-0.017 (0.035)
Treatment 2		0.115 (0.399)		-1.535** (0.667)		0.079** (0.033)
Best Price 2023 x T1		0.396 (0.444)		1.693*** (0.650)		-0.048 (0.040)
Best Price 2023 x T2		-0.244 (0.407)		1.428** (0.682)		-0.060 (0.039)
Observations	977	977	977	977	977	977

Note: Outcomes are ln(Harvest Cost, Farmer), the log of harvest-time costs borne by the farmer, and Harvest Cost Farmer's Share, the farmer-paid share of total harvest costs (farmer plus buyer). Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for whether the buyer offered the best price in 2023, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.10.

Table 102: Heterogeneous Treatment Effects: Additional Services Provided by Buyer, by whether the buyer offered the best price in 2023

	Additional Services 2024	
	(1)	(2)
Treated	-0.108*	
	(0.055)	
Treatment 1		-0.039
		(0.068)
Treatment 2		-0.143**
		(0.056)
Best Price 2023 x Treated	0.172***	
	(0.051)	
Best Price 2023 x T1		0.167***
		(0.058)
Best Price 2023 x T2		0.120**
		(0.057)
Observations	977	977

Note: Additional Services is a binary indicator equal to one if the buyer provided additional services (e.g. financial assistance or inputs) at cultivation. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for whether the buyer offered the best price in 2023, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 103: Heterogeneous Treatment Effects: Agents Considered and Buyer Search Time, by whether the buyer offered the best price in 2023

	Agents Considered 2024		Buyer Search Time (Days) 2024	
	(1)	(2)	(3)	(4)
Treated	-0.010 (0.242)		-3.174 (2.302)	
Treatment 1		0.220 (0.280)		-1.163 (2.748)
Treatment 2		-0.134 (0.263)		-4.266* (2.261)
Best Price 2023 x Treated	-0.162 (0.233)		2.912 (2.012)	
Best Price 2023 x T1		-0.305 (0.288)		1.113 (2.686)
Best Price 2023 x T2		-0.163 (0.227)		3.643** (1.852)
Observations	977	977	975	975

Note: Outcomes are the number of buyers/agents the farmer considered before selling, and the number of days taken to find a buyer and complete the sale. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for whether the buyer offered the best price in 2023, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 104: Heterogeneous Treatment Effects: Sowing Choice, New Agent Consideration and Credit Constraint, by whether the buyer offered the best price in 2023

	Plant Potato (Expectation 2025) (1)	(2)	New Agent (Expectation 2025) (3)	(4)	Credit Constraint (Expectation 2025) (5)	(6)
Treated	-0.050 (0.053)		-0.032 (0.068)		0.008 (0.035)	
Treatment 1		-0.087 (0.082)		-0.131 (0.106)		0.002 (0.038)
Treatment 2		-0.031 (0.052)		0.022 (0.065)		0.013 (0.040)
Best Price 2023 x Treated	0.034 (0.054)		0.048 (0.086)		0.100** (0.046)	
Best Price 2023 x T1		0.039 (0.081)		0.110 (0.120)		0.149*** (0.053)
Best Price 2023 x T2		0.058 (0.051)		0.049 (0.086)		0.038 (0.053)
Observations	977	977	976	976	977	977

Note: Outcomes are Plant Potato (Expectation 2025) (intention to cultivate potatoes again in 2025), New Agent (Expectation 2025) (intention to consider a new Commission Agent), and Credit Constraint (Expectation 2025) (self-reported credit constraint), all reported at the 2024 endline, not realised 2025 outcomes. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for whether the buyer offered the best price in 2023, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

3.13 Buyer Provided Additional Services in 2023

Table 105: Heterogeneous Treatment Effects: Sales, Quantities, Prices and Profits, by whether the buyer provided additional services in 2023

	ln(Sales_2024)		ln(Quantity_2024)		ln(Costs_2024)		ln(Price_2024)		ln(Profit_2024)	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Treated	0.042		0.062		-0.057		-0.015		0.057	
	(0.114)		(0.093)		(0.095)		(0.037)		(0.150)	
Treatment 1		0.046		0.049		-0.119		0.004		0.034
		(0.121)		(0.098)		(0.098)		(0.044)		(0.146)
Treatment 2		0.037		0.076		0.011		-0.037		0.081
		(0.141)		(0.109)		(0.108)		(0.043)		(0.214)
Services 2023 x Treated	-0.862**		-0.724**		-1.001***		0.183**		-0.774	
	(0.383)		(0.301)		(0.308)		(0.076)		(0.706)	
Services 2023 x T1		-0.810*		-0.750		-1.147***		0.225**		-0.442
		(0.489)		(0.464)		(0.431)		(0.096)		(0.755)
Services 2023 x T2		-0.892*		-0.715**		-0.943**		0.166**		-0.989
		(0.466)		(0.346)		(0.389)		(0.076)		(0.738)
Observations	977	977	977	977	977	977	977	977	977	977

Note: ln(Sales), ln(Quantity), ln(Costs), ln(Price) and ln(Profit) are the natural logs of total sales value, quantity sold, cultivation/harvest costs, unit price received, and profit, each measured per hectare of total cultivated area. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for whether the buyer provided additional services in 2023, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.10.

Table 106: Heterogeneous Treatment Effects: Percentage of Sales by Buyer Type, by whether the buyer provided additional services in 2023

	CA 2024		Informal Agent 2024		Other 2024	
	(1)	(2)	(3)	(4)	(5)	(6)
Treated	-0.018 (0.059)		0.032 (0.057)		-0.016 (0.016)	
Treatment 1		-0.052 (0.058)		0.052 (0.057)		-0.000 (0.019)
Treatment 2		0.019 (0.081)		0.011 (0.078)		-0.033* (0.017)
Services 2023 x Treated	-0.230** (0.107)		0.251*** (0.091)		-0.019 (0.052)	
Services 2023 x T1		-0.264** (0.103)		0.292** (0.126)		-0.024 (0.061)
Services 2023 x T2		-0.226* (0.119)		0.235** (0.092)		-0.008 (0.057)
Observations	977	977	977	977	977	977

Note: Outcomes are the share of the farmer's total sales value sold to the Commission Agent (CA), to an informal agent, and to other buyer types. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for whether the buyer provided additional services in 2023, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 107: Heterogeneous Treatment Effects: Flexible Arrangements, New Agent Consideration and Buyer Change, by whether the buyer provided additional services in 2023

	Flexible 2024		New Agent 2024		Change Buyer 2024	
	(1)	(2)	(3)	(4)	(5)	(6)
Treated	0.002 (0.053)		-0.156 (0.356)		-0.002 (0.062)	
Treatment 1		0.020 (0.057)		0.020 (0.388)		0.080 (0.059)
Treatment 2		-0.017 (0.064)		-0.353 (0.407)		-0.093 (0.080)
Services 2023 x Treated	-0.136** (0.068)		0.243 (0.932)		-0.168 (0.272)	
Services 2023 x T1		-0.105 (0.066)		1.494 (0.987)		0.078 (0.247)
Services 2023 x T2		-0.149* (0.081)		-0.434 (0.790)		-0.277 (0.264)
Observations	977	977	977	977	977	977

Note: Change Buyer is a binary indicator equal to one if the farmer sold to a different buyer/agent than at baseline. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for whether the buyer provided additional services in 2023, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.10.

Table 108: Heterogeneous Treatment Effects: Harvest Costs, by whether the buyer provided additional services in 2023

	ln(Harvest Cost Farmer '24) (1)	ln(Harvest Cost Buyer '24) (2)	ln(Harvest Cost Buyer '24) (3)	Harvest Cost Share '24 (4)	Harvest Cost Share '24 (5)	Harvest Cost Share '24 (6)
Treated	-0.226 (0.188)		0.198 (0.397)		-0.010 (0.023)	
Treatment 1		-0.344* (0.194)		0.915** (0.463)		-0.054** (0.025)
Treatment 2		-0.093 (0.254)		-0.586 (0.429)		0.037 (0.027)
Services 2023 x Treated	-0.617 (0.690)		-0.007 (1.098)		-0.010 (0.039)	
Services 2023 x T1		-1.229 (0.887)		-0.072 (1.216)		-0.021 (0.050)
Services 2023 x T2		-0.300 (0.631)		0.397 (1.147)		-0.026 (0.033)
Observations	977	977	977	977	977	977

Note: Outcomes are ln(Harvest Cost, Farmer), the log of harvest-time costs borne by the farmer, and Harvest Cost Farmer's Share, the farmer-paid share of total harvest costs (farmer plus buyer). Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for whether the buyer provided additional services in 2023, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.10.

Table 109: Heterogeneous Treatment Effects: Additional Services Provided by Buyer, by whether the buyer provided additional services in 2023

	Additional Services 2024	
	(1)	(2)
Treated	0.015 (0.034)	
Treatment 1		0.086** (0.040)
Treatment 2		-0.062* (0.036)
Services 2023 x Treated	0.030 (0.102)	
Services 2023 x T1		0.055 (0.102)
Services 2023 x T2		0.050 (0.108)
Observations	977	977

Note: Additional Services is a binary indicator equal to one if the buyer provided additional services (e.g. financial assistance or inputs) at cultivation. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for whether the buyer provided additional services in 2023, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 110: Heterogeneous Treatment Effects: Agents Considered and Buyer Search Time, by whether the buyer provided additional services in 2023

	Agents Considered 2024 (1)	(2)	Buyer Search Time (Days) 2024 (3)	(4)
Treated	-0.129 (0.181)		-1.225 (1.307)	
Treatment 1		-0.074 (0.200)		-0.722 (1.374)
Treatment 2		-0.192 (0.214)		-1.778 (1.455)
Services 2023 x Treated	0.193 (0.664)		1.409 (1.600)	
Services 2023 x T1		1.254** (0.593)		2.379 (1.723)
Services 2023 x T2		-0.430 (0.583)		1.071 (1.645)
Observations	977	977	975	975

Note: Outcomes are the number of buyers/agents the farmer considered before selling, and the number of days taken to find a buyer and complete the sale. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for whether the buyer provided additional services in 2023, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 111: Heterogeneous Treatment Effects: Sowing Choice, New Agent Consideration and Credit Constraint, by whether the buyer provided additional services in 2023

	Plant Potato (Expectation 2025) (1)	New Agent (Expectation 2025) (2)	Credit Constraint (Expectation 2025) (3)	(4)	(5)	(6)
Treated	-0.024 (0.039)		0.004 (0.047)		0.078*** (0.030)	
Treatment 1		-0.056 (0.049)		-0.042 (0.061)		0.118*** (0.031)
Treatment 2		0.011 (0.037)		0.054 (0.047)		0.035 (0.035)
Services 2023 x Treated	-0.077 (0.051)		-0.232** (0.108)		-0.018 (0.094)	
Services 2023 x T1		-0.061 (0.082)		-0.263* (0.141)		-0.070 (0.097)
Services 2023 x T2		-0.103* (0.055)		-0.237** (0.108)		0.035 (0.107)
Observations	977	977	976	976	977	977

Note: Outcomes are Plant Potato (Expectation 2025) (intention to cultivate potatoes again in 2025), New Agent (Expectation 2025) (intention to consider a new Commission Agent), and Credit Constraint (Expectation 2025) (self-reported credit constraint), all reported at the 2024 endline, not realised 2025 outcomes. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for whether the buyer provided additional services in 2023, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

3.14 Understanding With Buyer For Quantity To Be Sold In 2024

Table 112: Heterogeneous Treatment Effects: Sales, Quantities, Prices and Profits, by understanding with the buyer on quantity to be sold

	ln(Sales_2024)		ln(Quantity_2024)		ln(Costs_2024)		ln(Price_2024)		ln(Profit_2024)	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Treated	0.033		0.012		0.016		-0.043		-0.007	
	(0.213)		(0.224)		(0.157)		(0.091)		(0.219)	
Treatment 1		-0.037		-0.037		-0.086		-0.017		-0.011
		(0.209)		(0.223)		(0.151)		(0.094)		(0.210)
Treatment 2		0.193		0.127		0.262		-0.104		0.006
		(0.258)		(0.258)		(0.193)		(0.105)		(0.410)
Understanding x Treated	-0.019		0.033		-0.110		0.037		0.069	
	(0.211)		(0.215)		(0.162)		(0.094)		(0.236)	
Understanding x T1		0.077		0.079		-0.068		0.026		0.052
		(0.215)		(0.219)		(0.163)		(0.097)		(0.266)
Understanding x T2		-0.201		-0.080		-0.304		0.086		0.074
		(0.245)		(0.247)		(0.200)		(0.109)		(0.387)
Observations	977	977	977	977	977	977	977	977	977	977

Note: ln(Sales), ln(Quantity), ln(Costs), ln(Price) and ln(Profit) are the natural logs of total sales value, quantity sold, cultivation/harvest costs, unit price received, and profit, each measured per hectare of total cultivated area. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for whether the farmer had an understanding with the buyer on quantity to be sold in 2024, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.10.

Table 113: Heterogeneous Treatment Effects: Percentage of Sales by Buyer Type, by understanding with the buyer on quantity to be sold

	CA 2024		Informal Agent 2024		Other 2024	
	(1)	(2)	(3)	(4)	(5)	(6)
Treated	-0.237** (0.113)		0.200* (0.117)		0.033** (0.017)	
Treatment 1		-0.156 (0.109)		0.116 (0.113)		0.035* (0.020)
Treatment 2		-0.420*** (0.113)		0.392*** (0.128)		0.024 (0.025)
Understanding x Treated	0.264** (0.120)		-0.208* (0.119)		-0.054** (0.025)	
Understanding x T1		0.111 (0.109)		-0.072 (0.113)		-0.035 (0.028)
Understanding x T2		0.508*** (0.130)		-0.444*** (0.135)		-0.064* (0.034)
Observations	977	977	977	977	977	977

Note: Outcomes are the share of the farmer's total sales value sold to the Commission Agent (CA), to an informal agent, and to other buyer types. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for whether the farmer had an understanding with the buyer on quantity to be sold in 2024, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 114: Heterogeneous Treatment Effects: Flexible Arrangements, New Agent Consideration and Buyer Change, by understanding with the buyer on quantity to be sold

	Flexible 2024		New Agent 2024		Change Buyer 2024	
	(1)	(2)	(3)	(4)	(5)	(6)
Treated	-0.052 (0.072)		0.369 (0.670)		0.145* (0.078)	
Treatment 1		-0.021 (0.064)		0.307 (0.697)		0.178** (0.087)
Treatment 2		-0.126 (0.110)		0.464 (0.694)		0.054 (0.090)
Understanding x Treated	0.059 (0.076)		-0.617 (0.733)		-0.192* (0.114)	
Understanding x T1		0.036 (0.070)		-0.171 (0.788)		-0.115 (0.111)
Understanding x T2		0.125 (0.105)		-1.041 (0.759)		-0.195 (0.130)
Observations	977	977	977	977	977	977

Note: Change Buyer is a binary indicator equal to one if the farmer sold to a different buyer/agent than at baseline. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for whether the farmer had an understanding with the buyer on quantity to be sold in 2024, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 115: Heterogeneous Treatment Effects: Harvest Costs, by understanding with the buyer on quantity to be sold

	ln(Harvest Cost Farmer '24) (1)	ln(Harvest Cost Buyer '24) (2)	ln(Harvest Cost Buyer '24) (3)	ln(Harvest Cost Buyer '24) (4)	Harvest Cost Share '24 (5)	Harvest Cost Share '24 (6)
Treated	-0.963*** (0.359)		0.119 (0.977)		-0.047 (0.050)	
Treatment 1		-0.730** (0.287)		0.556 (1.018)		-0.059 (0.053)
Treatment 2		-1.472** (0.699)		-1.005 (0.996)		-0.014 (0.060)
Understanding x Treated	0.886** (0.445)		0.052 (0.984)		0.047 (0.057)	
Understanding x T1		0.369 (0.363)		0.429 (1.000)		0.003 (0.055)
Understanding x T2		1.644** (0.804)		0.479 (1.027)		0.061 (0.069)
Observations	977	977	977	977	977	977

Note: Outcomes are ln(Harvest Cost, Farmer), the log of harvest-time costs borne by the farmer, and Harvest Cost Farmer's Share, the farmer-paid share of total harvest costs (farmer plus buyer). Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for whether the farmer had an understanding with the buyer on quantity to be sold in 2024, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.10.

Table 116: Heterogeneous Treatment Effects: Additional Services Provided by Buyer, by understanding with the buyer on quantity to be sold

	Additional Services 2024	
	(1)	(2)
Treated	-0.006 (0.099)	
Treatment 1		0.021 (0.106)
Treatment 2		-0.080 (0.100)
Understanding x Treated	0.022 (0.100)	
Understanding x T1		0.087 (0.103)
Understanding x T2		0.018 (0.104)
Observations	977	977

Note: Additional Services is a binary indicator equal to one if the buyer provided additional services (e.g. financial assistance or inputs) at cultivation. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for whether the farmer had an understanding with the buyer on quantity to be sold in 2024, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 117: Heterogeneous Treatment Effects: Agents Considered and Buyer Search Time, by understanding with the buyer on quantity to be sold

	Agents Considered 2024		Buyer Search Time (Days) 2024	
	(1)	(2)	(3)	(4)
Treated	0.645*** (0.238)		2.283 (2.887)	
Treatment 1		0.559** (0.252)		1.343 (2.916)
Treatment 2		0.824*** (0.256)		4.325 (3.289)
Understanding x Treated	-0.905*** (0.300)		-4.358 (3.018)	
Understanding x T1		-0.653** (0.284)		-2.431 (2.937)
Understanding x T2		-1.226*** (0.357)		-7.253** (3.492)
Observations	977	977	975	975

Note: Outcomes are the number of buyers/agents the farmer considered before selling, and the number of days taken to find a buyer and complete the sale. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for whether the farmer had an understanding with the buyer on quantity to be sold in 2024, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 118: Heterogeneous Treatment Effects: Sowing Choice, New Agent Consideration and Credit Constraint, by understanding with the buyer on quantity to be sold

	Plant Potato (Expectation 2025) (1)	New Agent (Expectation 2025) (2)	(3)	(4)	Credit Constraint (Expectation 2025) (5)	(6)
Treated	-0.022 (0.078)		-0.123 (0.112)		0.081 (0.055)	
Treatment 1		-0.039 (0.085)		-0.146 (0.113)		0.101* (0.058)
Treatment 2		0.023 (0.069)		-0.065 (0.121)		0.031 (0.062)
Understanding x Treated	-0.003 (0.065)		0.150 (0.112)		-0.008 (0.066)	
Understanding x T1		-0.023 (0.068)		0.120 (0.114)		0.013 (0.067)
Understanding x T2		-0.016 (0.065)		0.136 (0.122)		0.009 (0.082)
Observations	977	977	976	976	977	977

Note: Outcomes are Plant Potato (Expectation 2025) (intention to cultivate potatoes again in 2025), New Agent (Expectation 2025) (intention to consider a new Commission Agent), and Credit Constraint (Expectation 2025) (self-reported credit constraint), all reported at the 2024 endline, not realised 2025 outcomes. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for whether the farmer had an understanding with the buyer on quantity to be sold in 2024, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

3.15 Potential Buyer Provided Financial Assistance at Cultivation

Table 119: Heterogeneous Treatment Effects: Sales, Quantities, Prices and Profits, by whether a potential buyer offered financial assistance at cultivation

	ln(Sales_2024)		ln(Quantity_2024)		ln(Costs_2024)		ln(Price_2024)		ln(Profit_2024)	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Treated	0.034 (0.118)		0.040 (0.099)		-0.060 (0.098)		-0.008 (0.036)		0.043 (0.156)	
Treatment 1		0.026 (0.126)		0.015 (0.103)		-0.144 (0.099)		0.013 (0.044)		0.004 (0.154)
Treatment 2		0.035 (0.140)		0.060 (0.112)		0.021 (0.112)		-0.030 (0.042)		0.068 (0.207)
Cultivation Services x Treated	0.419 (0.608)		0.519 (0.664)		0.053 (0.619)		0.153 (0.093)		1.231 (0.942)	
Cultivation Services x T1		1.042 (0.724)		1.231* (0.706)		0.737 (0.752)		0.098 (0.096)		2.617*** (0.824)
Cultivation Services x T2		-0.339 (0.345)		-0.352 (0.413)		-0.786*** (0.254)		0.219** (0.087)		-0.450 (0.424)
Observations	977	977	977	977	977	977	977	977	977	977

Note: ln(Sales), ln(Quantity), ln(Costs), ln(Price) and ln(Profit) are the natural logs of total sales value, quantity sold, cultivation/harvest costs, unit price received, and profit, each measured per hectare of total cultivated area. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for whether a potential buyer offered financial assistance at cultivation, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.10.

Table 120: Heterogeneous Treatment Effects: Percentage of Sales by Buyer Type, by whether a potential buyer offered financial assistance at cultivation

	CA 2024		Informal Agent 2024		Other 2024	
	(1)	(2)	(3)	(4)	(5)	(6)
Treated	-0.024 (0.063)		0.037 (0.060)		-0.016 (0.016)	
Treatment 1		-0.068 (0.060)		0.067 (0.059)		0.000 (0.019)
Treatment 2		0.018 (0.086)		0.010 (0.082)		-0.032* (0.017)
Cultivation Services x Treated	0.276 (0.308)		-0.290 (0.263)		0.018 (0.061)	
Cultivation Services x T1		0.728*** (0.219)		-0.682*** (0.189)		-0.041 (0.056)
Cultivation Services x T2		-0.278 (0.192)		0.190 (0.155)		0.089 (0.055)
Observations	977	977	977	977	977	977

Note: Outcomes are the share of the farmer's total sales value sold to the Commission Agent (CA), to an informal agent, and to other buyer types. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for whether a potential buyer offered financial assistance at cultivation, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 121: Heterogeneous Treatment Effects: Flexible Arrangements, New Agent Consideration and Buyer Change, by whether a potential buyer offered financial assistance at cultivation

	Flexible 2024		New Agent 2024		Change Buyer 2024	
	(1)	(2)	(3)	(4)	(5)	(6)
Treated	0.007 (0.053)		-0.214 (0.366)		-0.018 (0.065)	
Treatment 1		0.026 (0.058)		0.043 (0.396)		0.084 (0.061)
Treatment 2		-0.011 (0.064)		-0.483 (0.433)		-0.122 (0.087)
Cultivation Services x Treated	-0.094 (0.108)		0.338 (0.713)		0.186 (0.319)	
Cultivation Services x T1		-0.232** (0.100)		0.288 (0.777)		-0.157 (0.394)
Cultivation Services x T2		0.071 (0.073)		0.390 (0.768)		0.602*** (0.222)
Observations	977	977	977	977	977	977

Note: Change Buyer is a binary indicator equal to one if the farmer sold to a different buyer/agent than at baseline. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for whether a potential buyer offered financial assistance at cultivation, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 122: Heterogeneous Treatment Effects: Harvest Costs, by whether a potential buyer offered financial assistance at cultivation

	ln(Harvest Cost Farmer '24) (1)	ln(Harvest Cost Buyer '24) (2)	ln(Harvest Cost Buyer '24) (3)	ln(Harvest Cost Buyer '24) (4)	Harvest Cost Share '24 (5)	Harvest Cost Share '24 (6)
Treated	-0.226 (0.199)		0.084 (0.413)		-0.009 (0.023)	
Treatment 1		-0.399* (0.208)		0.869* (0.480)		-0.055** (0.025)
Treatment 2		-0.048 (0.261)		-0.689 (0.435)		0.039 (0.026)
Cultivation Services x Treated	0.339 (0.741)		5.439** (2.704)		-0.027 (0.062)	
Cultivation Services x T1		0.739 (0.960)		0.713 (0.686)		0.051 (0.055)
Cultivation Services x T2		-0.147 (0.713)		11.235*** (0.480)		-0.120** (0.058)
Observations	977	977	977	977	977	977

Note: Outcomes are ln(Harvest Cost, Farmer), the log of harvest-time costs borne by the farmer, and Harvest Cost Farmer's Share, the farmer-paid share of total harvest costs (farmer plus buyer). Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for whether a potential buyer offered financial assistance at cultivation, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.10.

Table 123: Heterogeneous Treatment Effects: Additional Services Provided by Buyer, by whether a potential buyer offered financial assistance at cultivation

	Additional Services 2024	
	(1)	(2)
Treated	0.004 (0.035)	
Treatment 1		0.082** (0.039)
Treatment 2		-0.074** (0.037)
Cultivation Services x Treated	0.522** (0.248)	
Cultivation Services x T1		0.084 (0.067)
Cultivation Services x T2		1.054*** (0.048)
Observations	977	977

Note: Additional Services is a binary indicator equal to one if the buyer provided additional services (e.g. financial assistance or inputs) at cultivation. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for whether a potential buyer offered financial assistance at cultivation, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 124: Heterogeneous Treatment Effects: Agents Considered and Buyer Search Time, by whether a potential buyer offered financial assistance at cultivation

	Agents Considered 2024		Buyer Search Time (Days) 2024	
	(1)	(2)	(3)	(4)
Treated	-0.122 (0.185)		-1.479 (1.263)	
Treatment 1		-0.007 (0.201)		-0.848 (1.315)
Treatment 2		-0.239 (0.230)		-2.144 (1.425)
Cultivation Services x Treated	-1.253** (0.542)		3.336* (1.825)	
Cultivation Services x T1		-1.551** (0.604)		3.678 (2.247)
Cultivation Services x T2		-0.892 (0.545)		2.904* (1.536)
Observations	977	977	975	975

Note: Outcomes are the number of buyers/agents the farmer considered before selling, and the number of days taken to find a buyer and complete the sale. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for whether a potential buyer offered financial assistance at cultivation, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.10.

Table 125: Heterogeneous Treatment Effects: Sowing Choice, New Agent Consideration and Credit Constraint, by whether a potential buyer offered financial assistance at cultivation

	Plant Potato (Expectation 2025) (1)	New Agent (Expectation 2025) (2)	New Agent (Expectation 2025) (3)	Credit Constraint (Expectation 2025) (4)	Credit Constraint (Expectation 2025) (5)	Credit Constraint (Expectation 2025) (6)
Treated	-0.028 (0.039)		0.002 (0.048)		0.072** (0.031)	
Treatment 1		-0.064 (0.049)		-0.055 (0.062)		0.115*** (0.031)
Treatment 2		0.007 (0.037)		0.058 (0.047)		0.033 (0.036)
Cultivation Services x Treated	0.028 (0.216)		0.020 (0.323)		0.391 (0.276)	
Cultivation Services x T1		0.327** (0.166)		0.549*** (0.189)		-0.067 (0.056)
Cultivation Services x T2		-0.339*** (0.118)		-0.627*** (0.104)		0.951*** (0.038)
Observations	977	977	976	976	977	977

Note: Outcomes are Plant Potato (Expectation 2025) (intention to cultivate potatoes again in 2025), New Agent (Expectation 2025) (intention to consider a new Commission Agent), and Credit Constraint (Expectation 2025) (self-reported credit constraint), all reported at the 2024 endline, not realised 2025 outcomes. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for whether a potential buyer offered financial assistance at cultivation, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

3.16 Buyers Considered Before Selling

Table 126: Heterogeneous Treatment Effects: Sales, Quantities, Prices and Profits, by the number of buyers considered

	(1) ln(Sales_2024)	(2) ln(Sales_2024)	(3) ln(Quantity_2024)	(4) ln(Quantity_2024)	(5) ln(Costs_2024)	(6) ln(Costs_2024)	(7) ln(Price_2024)	(8) ln(Price_2024)	(9) ln(Profit_2024)	(10) ln(Profit_2024)
Treated	-0.006 (0.147)		-0.021 (0.121)		-0.037 (0.132)		0.012 (0.046)		-0.095 (0.136)	
High Agent Count x Treated	0.055 (0.167)		0.122 (0.133)		-0.125 (0.150)		-0.038 (0.058)		0.299 (0.237)	
Treatment 1		-0.056 (0.158)		-0.072 (0.123)		-0.161 (0.140)		0.018 (0.056)		-0.181 (0.187)
Treatment 2		0.039 (0.158)		0.024 (0.138)		0.068 (0.146)		0.009 (0.051)		-0.020 (0.172)
High Agent Count x T1		0.160 (0.168)		0.196 (0.137)		-0.011 (0.149)		-0.004 (0.059)		0.424 (0.294)
High Agent Count x T2		-0.060 (0.242)		0.050 (0.180)		-0.208 (0.206)		-0.087 (0.082)		0.178 (0.272)
Observations	977	977	977	977	977	977	977	977	977	977

Note: ln(Sales), ln(Quantity), ln(Costs), ln(Price) and ln(Profit) are the natural logs of total sales value, quantity sold, cultivation/harvest costs, unit price received, and profit, each measured per hectare of total cultivated area. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for the number of buyers considered before selling, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.10.

Table 127: Heterogeneous Treatment Effects: Percentage of Sales by Buyer Type, by the number of buyers considered

	CA 2024 (1)	(2)	Informal Agent 2024 (3)	(4)	Other 2024 (5)	(6)
Treated	-0.058 (0.089)		0.049 (0.085)		0.005 (0.014)	
Treatment 1		-0.108 (0.083)		0.089 (0.082)		0.016 (0.020)
Treatment 2		-0.017 (0.113)		0.015 (0.108)		-0.004 (0.017)
High Agent Count x Treated	0.065 (0.086)		-0.016 (0.085)		-0.044* (0.023)	
High Agent Count x T1		0.089 (0.075)		-0.051 (0.078)		-0.033 (0.026)
High Agent Count x T2		0.063 (0.123)		0.008 (0.116)		-0.066*** (0.024)
Observations	977	977	977	977	977	977

Note: Outcomes are the share of the farmer's total sales value sold to the Commission Agent (CA), to an informal agent, and to other buyer types. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for the number of buyers considered before selling, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 128: Heterogeneous Treatment Effects: Flexible Arrangements, New Agent Consideration and Buyer Change, by the number of buyers considered

	Flexible 2024		New Agent 2024		Change Buyer 2024	
	(1)	(2)	(3)	(4)	(5)	(6)
Treated	0.026 (0.068)		-0.323 (0.506)		-0.042 (0.099)	
Treatment 1		0.051 (0.072)		0.210 (0.590)		0.101 (0.089)
Treatment 2		0.005 (0.078)		-0.784 (0.523)		-0.163 (0.113)
High Agent Count x Treated	-0.053 (0.062)		0.362 (0.500)		0.074 (0.098)	
High Agent Count x T1		-0.069 (0.064)		-0.278 (0.646)		-0.031 (0.092)
High Agent Count x T2		-0.045 (0.071)		0.929* (0.498)		0.135 (0.107)
Observations	977	977	977	977	977	977

Note: Change Buyer is a binary indicator equal to one if the farmer sold to a different buyer/agent than at baseline. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for the number of buyers considered before selling, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.10.

Table 129: Heterogeneous Treatment Effects: Harvest Costs, by the number of buyers considered

	(1) ln(Harvest Cost Farmer '24)	(2) ln(Harvest Cost Farmer '24)	(3) ln(Harvest Cost Buyer '24)	(4) ln(Harvest Cost Buyer '24)	(5) Harvest Cost Share '24	(6) Harvest Cost Share '24
Treated	-0.356 (0.290)		-0.104 (0.543)		-0.010 (0.028)	
High Agent Count x Treated	0.237 (0.325)		0.622 (0.607)		-0.000 (0.031)	
Treatment 1		-0.541* (0.292)		0.753 (0.598)		-0.064** (0.030)
Treatment 2		-0.201 (0.367)		-0.818 (0.560)		0.034 (0.030)
High Agent Count x T1		0.273 (0.392)		0.307 (0.685)		0.018 (0.038)
High Agent Count x T2		0.299 (0.369)		0.535 (0.559)		0.008 (0.029)
Observations	977	977	977	977	977	977

Note: Outcomes are ln(Harvest Cost, Farmer), the log of harvest-time costs borne by the farmer, and Harvest Cost Farmer's Share, the farmer-paid share of total harvest costs (farmer plus buyer). Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for the number of buyers considered before selling, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.10.

Table 130: Heterogeneous Treatment Effects: Additional Services Provided by Buyer, by the number of buyers considered

	Additional Services 2024	
	(1)	(2)
Treated	-0.011 (0.047)	
Treatment 1		0.076 (0.052)
Treatment 2		-0.084* (0.047)
High Agent Count x Treated	0.053 (0.050)	
High Agent Count x T1		0.021 (0.055)
High Agent Count x T2		0.045 (0.048)
Observations	977	977

Note: Additional Services is a binary indicator equal to one if the buyer provided additional services (e.g. financial assistance or inputs) at cultivation. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for the number of buyers considered before selling, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 131: Heterogeneous Treatment Effects: Agents Considered and Buyer Search Time, by the number of buyers considered

	Agents Considered 2024		Buyer Search Time (Days) 2024	
	(1)	(2)	(3)	(4)
Treated	0.090 (0.216)		-1.515 (1.701)	
Treatment 1		0.289 (0.212)		-0.546 (1.767)
Treatment 2		-0.080 (0.272)		-2.343 (1.812)
High Agent Count x Treated	-0.388 (0.247)		0.864 (2.141)	
High Agent Count x T1		-0.591** (0.239)		0.002 (2.297)
High Agent Count x T2		-0.225 (0.280)		1.478 (2.549)
Observations	977	977	975	975

Note: Outcomes are the number of buyers/agents the farmer considered before selling, and the number of days taken to find a buyer and complete the sale. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for the number of buyers considered before selling, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 132: Heterogeneous Treatment Effects: Sowing Choice, New Agent Consideration and Credit Constraint, by the number of buyers considered

	Plant Potato (Expectation 2025)		New Agent (Expectation 2025)		Credit Constraint (Expectation 2025)	
	(1)	(2)	(3)	(4)	(5)	(6)
Treated	-0.014 (0.047)		-0.000 (0.064)		0.053 (0.035)	
Treatment 1		-0.057 (0.054)		-0.085 (0.080)		0.094** (0.039)
Treatment 2		0.022 (0.048)		0.072 (0.062)		0.018 (0.040)
High Agent Count x Treated	-0.026 (0.049)		-0.004 (0.070)		0.052 (0.036)	
High Agent Count x T1		-0.001 (0.054)		0.071 (0.080)		0.040 (0.044)
High Agent Count x T2		-0.035 (0.053)		-0.058 (0.084)		0.044 (0.041)
Observations	977	977	976	976	977	977

Note: Outcomes are Plant Potato (Expectation 2025) (intention to cultivate potatoes again in 2025), New Agent (Expectation 2025) (intention to consider a new Commission Agent), and Credit Constraint (Expectation 2025) (self-reported credit constraint), all reported at the 2024 endline, not realised 2025 outcomes. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for the number of buyers considered before selling, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

3.17 Market Awareness

Table 133: Heterogeneous Treatment Effects: Sales, Quantities, Prices and Profits, by market awareness

	(1) ln(Sales_2024)	(2) ln(Sales_2024)	(3) ln(Quantity_2024)	(4) ln(Quantity_2024)	(5) ln(Costs_2024)	(6) ln(Costs_2024)	(7) ln(Price_2024)	(8) ln(Price_2024)	(9) ln(Profit_2024)	(10) ln(Profit_2024)
Treated	0.057 (0.132)		0.061 (0.101)		-0.035 (0.097)		0.010 (0.039)		0.052 (0.156)	
High Mkt Awareness x Treated	-0.121 (0.185)		-0.073 (0.179)		-0.201 (0.178)		-0.050 (0.048)		-0.018 (0.239)	
Treatment 1		0.050 (0.142)		0.023 (0.109)		-0.101 (0.108)		0.028 (0.048)		0.040 (0.177)
Treatment 2		0.063 (0.155)		0.098 (0.112)		0.029 (0.109)		-0.008 (0.048)		0.066 (0.207)
High Mkt Awareness x T1		-0.130 (0.184)		-0.007 (0.196)		-0.234 (0.184)		-0.061 (0.057)		-0.073 (0.292)
High Mkt Awareness x T2		-0.104 (0.237)		-0.161 (0.201)		-0.119 (0.220)		-0.040 (0.058)		0.072 (0.248)
Observations	977	977	977	977	977	977	977	977	977	977

Note: ln(Sales), ln(Quantity), ln(Costs), ln(Price) and ln(Profit) are the natural logs of total sales value, quantity sold, cultivation/harvest costs, unit price received, and profit, each measured per hectare of total cultivated area. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for baseline market awareness, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.10.

Table 134: Heterogeneous Treatment Effects: Percentage of Sales by Buyer Type, by market awareness

	CA 2024		Informal Agent 2024		Other 2024	
	(1)	(2)	(3)	(4)	(5)	(6)
Treated	0.005 (0.065)		0.005 (0.064)		-0.012 (0.018)	
Treatment 1		-0.042 (0.064)		0.043 (0.066)		-0.000 (0.020)
Treatment 2		0.052 (0.087)		-0.031 (0.085)		-0.023 (0.020)
High Mkt Awareness x Treated	-0.121* (0.068)		0.131* (0.076)		-0.013 (0.029)	
High Mkt Awareness x T1		-0.075 (0.083)		0.077 (0.088)		-0.005 (0.033)
High Mkt Awareness x T2		-0.175*** (0.067)		0.201** (0.081)		-0.029 (0.030)
Observations	977	977	977	977	977	977

Note: Outcomes are the share of the farmer's total sales value sold to the Commission Agent (CA), to an informal agent, and to other buyer types. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for baseline market awareness, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.10.

Table 135: Heterogeneous Treatment Effects: Flexible Arrangements, New Agent Consideration and Buyer Change, by market awareness

	Flexible 2024 (1)	(2)	New Agent 2024 (3)	(4)	Change Buyer 2024 (5)	(6)
Treated	-0.014 (0.052)		-0.199 (0.389)		0.002 (0.068)	
Treatment 1		0.011 (0.056)		0.046 (0.455)		0.112* (0.066)
Treatment 2		-0.038 (0.061)		-0.439 (0.451)		-0.105 (0.086)
High Mkt Awareness x Treated	0.056 (0.078)		0.116 (0.486)		-0.038 (0.090)	
High Mkt Awareness x T1		0.025 (0.081)		0.165 (0.617)		-0.096 (0.097)
High Mkt Awareness x T2		0.094 (0.094)		-0.071 (0.442)		0.003 (0.098)
Observations	977	977	977	977	977	977

Note: Change Buyer is a binary indicator equal to one if the farmer sold to a different buyer/agent than at baseline. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for baseline market awareness, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 136: Heterogeneous Treatment Effects: Harvest Costs, by market awareness

	(1)	(2)	(3)	(4)	(5)	(6)
	ln(Harvest Cost Farmer '24)	ln(Harvest Cost Farmer '24)	ln(Harvest Cost Buyer '24)	ln(Harvest Cost Buyer '24)	Harvest Cost Share '24	Harvest Cost Share '24
Treated	-0.232 (0.211)		-0.116 (0.436)		-0.001 (0.025)	
High Mkt Awareness x Treated	0.026 (0.327)		1.240** (0.531)		-0.037 (0.030)	
Treatment 1		-0.397* (0.235)		0.679 (0.480)		-0.050** (0.025)
Treatment 2		-0.070 (0.273)		-0.890* (0.460)		0.047* (0.028)
High Mkt Awareness x T1		-0.000 (0.442)		0.900 (0.632)		-0.019 (0.039)
High Mkt Awareness x T2		0.142 (0.321)		1.414** (0.709)		-0.043 (0.033)
Observations	977	977	977	977	977	977

Note: Outcomes are ln(Harvest Cost, Farmer), the log of harvest-time costs borne by the farmer, and Harvest Cost Farmer's Share, the farmer-paid share of total harvest costs (farmer plus buyer). Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for baseline market awareness, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.10.

Table 137: Heterogeneous Treatment Effects: Additional Services Provided by Buyer, by market awareness

	Additional Services 2024	
	(1)	(2)
Treated	-0.000 (0.038)	
Treatment 1		0.085** (0.039)
Treatment 2		-0.083** (0.040)
High Mkt Awareness x Treated	0.059 (0.055)	
High Mkt Awareness x T1		0.013 (0.063)
High Mkt Awareness x T2		0.093 (0.069)
Observations	977	977

Note: Additional Services is a binary indicator equal to one if the buyer provided additional services (e.g. financial assistance or inputs) at cultivation. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for baseline market awareness, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 138: Heterogeneous Treatment Effects: Agents Considered and Buyer Search Time, by market awareness

	Agents Considered 2024		Buyer Search Time (Days) 2024	
	(1)	(2)	(3)	(4)
Treated	-0.187 (0.194)		-1.362 (1.633)	
Treatment 1		-0.083 (0.218)		-0.651 (1.676)
Treatment 2		-0.289 (0.232)		-2.058 (1.798)
High Mkt Awareness x Treated	0.271 (0.192)		0.363 (2.658)	
High Mkt Awareness x T1		0.249 (0.230)		0.166 (2.940)
High Mkt Awareness x T2		0.257 (0.239)		0.355 (2.517)
Observations	977	977	975	975

Note: Outcomes are the number of buyers/agents the farmer considered before selling, and the number of days taken to find a buyer and complete the sale. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for baseline market awareness, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 139: Heterogeneous Treatment Effects: Sowing Choice, New Agent Consideration and Credit Constraint, by market awareness

	Plant Potato (Expectation 2025)		New Agent (Expectation 2025)		Credit Constraint (Expectation 2025)	
	(1)	(2)	(3)	(4)	(5)	(6)
Treated	-0.043 (0.034)		0.022 (0.049)		0.078*** (0.029)	
Treatment 1		-0.078* (0.045)		-0.019 (0.062)		0.105*** (0.031)
Treatment 2		-0.009 (0.033)		0.064 (0.051)		0.051 (0.035)
High Mkt Awareness x Treated	0.076 (0.047)		-0.101 (0.095)		0.005 (0.044)	
High Mkt Awareness x T1		0.075 (0.051)		-0.121 (0.113)		0.034 (0.055)
High Mkt Awareness x T2		0.093* (0.051)		-0.052 (0.090)		-0.053 (0.043)
Observations	977	977	976	976	977	977

Note: Outcomes are Plant Potato (Expectation 2025) (intention to cultivate potatoes again in 2025), New Agent (Expectation 2025) (intention to consider a new Commission Agent), and Credit Constraint (Expectation 2025) (self-reported credit constraint), all reported at the 2024 endline, not realised 2025 outcomes. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for baseline market awareness, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.10.

3.18 Expected Time to Sale

The pre-specified concept for this dimension is the time available, from the baseline survey, for the intervention to affect the farmer's sale (PAP, Followup 2, item 18). This is constructed from the farmer's baseline-reported expected sale date (not date of harvest, which is a distinct event in the data), as the gap between the baseline survey date and the earliest such date reported; the raw variable is winsorized at the 5th/95th percentile to bound a small number of implausible outliers before the sample is split at the median.

Table 140: Heterogeneous Treatment Effects: Sales, Quantities, Prices and Profits, by expected time to sale

Table 141: Heterogeneous Treatment Effects on Sales, Quantity Sold, Costs, Prices and Profits by Expected Time to Sale

	ln(Sales_2024)		ln(Quantity_2024)		ln(Costs_2024)		ln(Price_2024)		ln(Profit_2024)	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Treated	0.003 (0.150)		0.043 (0.122)		-0.042 (0.126)		-0.025 (0.043)		-0.031 (0.180)	
Treatment 1		-0.010 (0.173)		0.035 (0.125)		-0.106 (0.135)		-0.035 (0.060)		-0.103 (0.230)
Treatment 2		0.016 (0.166)		0.050 (0.139)		0.016 (0.138)		-0.017 (0.044)		0.034 (0.228)
High Time to Sale x Treated	0.032 (0.164)		-0.005 (0.122)		-0.091 (0.127)		0.036 (0.076)		0.153 (0.209)	
High Time to Sale x T1		0.047 (0.194)		-0.025 (0.130)		-0.119 (0.145)		0.089 (0.087)		0.227 (0.310)
High Time to Sale x T2		0.019 (0.186)		0.023 (0.147)		-0.038 (0.144)		-0.024 (0.078)		0.087 (0.183)
Observations	977	977	977	977	977	977	977	977	977	977

Note: ln(Sales), ln(Quantity), ln(Costs), ln(Price) and ln(Profit) are the natural logs of total sales value, quantity sold, cultivation/harvest costs, unit price received, and profit, each measured per hectare of total cultivated area. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for (winsorized) baseline expected time to sale, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.10.

Table 142: Heterogeneous Treatment Effects: Percentage of Sales by Buyer Type, by expected time to sale

Table 143: Heterogeneous Treatment Effects on Percentage Sales to Different Buyer Types by High_time_to_sale

	CA 2024		Informal Agent 2024		Other 2024	
	(1)	(2)	(3)	(4)	(5)	(6)
Treated	0.041 (0.081)		-0.007 (0.076)		-0.035 (0.023)	
Treatment 1		-0.027 (0.082)		0.050 (0.079)		-0.020 (0.023)
Treatment 2		0.104 (0.092)		-0.058 (0.087)		-0.048* (0.024)
High Time to Sale x Treated	-0.130* (0.076)		0.089 (0.078)		0.038 (0.033)	
High Time to Sale x T1		-0.069 (0.077)		0.025 (0.078)		0.037 (0.037)
High Time to Sale x T2		-0.185** (0.088)		0.149 (0.095)		0.034 (0.034)
Observations	977	977	977	977	977	977

Note: Outcomes are the share of the farmer's total sales value sold to the Commission Agent (CA), to an informal agent, and to other buyer types. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for (winsorized) baseline expected time to sale, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 144: Heterogeneous Treatment Effects: Flexible Arrangements, New Agent Consideration and Buyer Change, by expected time to sale

Table 145: Heterogeneous Treatment Effects on Flexible Arrangemenets, New Agents Considered and Changed Buyer by High_time_to_sale

	Flexible 2024		New Agent 2024		Change Buyer 2024	
	(1)	(2)	(3)	(4)	(5)	(6)
Treated	0.021 (0.071)		-0.085 (0.516)		0.002 (0.090)	
Treatment 1		0.043 (0.073)		0.162 (0.614)		0.098 (0.085)
Treatment 2		0.001 (0.081)		-0.312 (0.587)		-0.086 (0.110)
High Time to Sale x Treated	-0.043 (0.059)		-0.170 (0.477)		-0.029 (0.075)	
High Time to Sale x T1		-0.053 (0.066)		-0.133 (0.671)		-0.032 (0.080)
High Time to Sale x T2		-0.039 (0.071)		-0.293 (0.461)		-0.056 (0.083)
Observations	977	977	977	977	977	977

Note: Change Buyer is a binary indicator equal to one if the farmer sold to a different buyer/agent than at baseline. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for (winsorized) baseline expected time to sale, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 146: Heterogeneous Treatment Effects: Harvest Costs, by expected time to sale

Table 147: Heterogeneous Treatment Effects on Costs At Harvest by Expected Time to Sale

	ln(Harvest Cost Farmer '24) (1)	ln(Harvest Cost Buyer '24) (2)	ln(Harvest Cost Buyer '24) (3)	Harvest Cost Share '24 (4)	Harvest Cost Share '24 (5)	Harvest Cost Share '24 (6)
Treated	-0.114 (0.278)		-0.482 (0.472)		0.021 (0.027)	
Treatment 1		-0.361 (0.300)		0.317 (0.589)		-0.036 (0.030)
Treatment 2		0.111 (0.301)		-1.209*** (0.430)		0.073*** (0.024)
High Time to Sale x Treated	-0.240 (0.300)		1.378*** (0.506)		-0.063** (0.030)	
High Time to Sale x T1		-0.083 (0.317)		1.187* (0.647)		-0.038 (0.040)
High Time to Sale x T2		-0.349 (0.395)		1.354** (0.561)		-0.075** (0.034)
Observations	977	977	977	977	977	977

Note: Outcomes are ln(Harvest Cost, Farmer), the log of harvest-time costs borne by the farmer, and Harvest Cost Farmer's Share, the farmer-paid share of total harvest costs (farmer plus buyer). Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for (winsorized) baseline expected time to sale, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.10.

Table 148: Heterogeneous Treatment Effects: Additional Services Provided by Buyer, by expected time to sale

Table 149: Heterogeneous Treatment Effects on Provision of Additional Services by High_time_to_sale

	Additional Services 2024	
	(1)	(2)
Treated	-0.055 (0.045)	
Treatment 1		0.023 (0.055)
Treatment 2		-0.127*** (0.040)
High Time to Sale x Treated	0.144*** (0.052)	
High Time to Sale x T1		0.129* (0.067)
High Time to Sale x T2		0.137** (0.054)
Observations	977	977

Note: Additional Services is a binary indicator equal to one if the buyer provided additional services (e.g. financial assistance or inputs) at cultivation. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for (winsorized) baseline expected time to sale, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 150: Heterogeneous Treatment Effects: Agents Considered and Buyer Search Time, by expected time to sale

Table 151: Heterogeneous Treatment Effects on Agents Considered and Days Taken to Find a Buyer 2024 by High_time_to_sale

	Agents Considered 2024		Buyer Search Time (Days) 2024	
	(1)	(2)	(3)	(4)
Treated	-0.126 (0.214)		-0.829 (1.324)	
Treatment 1		0.027 (0.220)		0.695 (1.407)
Treatment 2		-0.264 (0.263)		-2.202 (1.389)
High Time to Sale x Treated	-0.006 (0.206)		-0.891 (1.367)	
High Time to Sale x T1		-0.089 (0.219)		-2.544 (1.562)
High Time to Sale x T2		0.045 (0.245)		0.612 (1.309)
Observations	977	977	975	975

Note: Outcomes are the number of buyers/agents the farmer considered before selling, and the number of days taken to find a buyer and complete the sale. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for (winsorized) baseline expected time to sale, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 152: Heterogeneous Treatment Effects: Sowing Choice, New Agent Consideration and Credit Constraint, by expected time to sale

Table 153: Heterogeneous Treatment Effects on Sowing Choice, Selling to New CA and Credit Constraint 2025 by High_time_to_sale

	Plant Potato (Expectation 2025)		New Agent (Expectation 2025)		Credit Constraint (Expectation 2025)	
	(1)	(2)	(3)	(4)	(5)	(6)
Treated	-0.035 (0.040)		0.012 (0.060)		0.070* (0.037)	
Treatment 1		-0.088* (0.046)		-0.044 (0.073)		0.108*** (0.041)
Treatment 2		0.013 (0.042)		0.063 (0.063)		0.035 (0.043)
High Time to Sale x Treated	0.020 (0.040)		-0.025 (0.079)		0.017 (0.039)	
High Time to Sale x T1		0.059 (0.042)		-0.012 (0.093)		0.013 (0.050)
High Time to Sale x T2		-0.010 (0.051)		-0.024 (0.080)		0.010 (0.050)
Observations	977	977	976	976	977	977

Note: Outcomes are Plant Potato (Expectation 2025) (intention to cultivate potatoes again in 2025), New Agent (Expectation 2025) (intention to consider a new Commission Agent), and Credit Constraint (Expectation 2025) (self-reported credit constraint), all reported at the 2024 endline, not realised 2025 outcomes. Treated is a pooled indicator for assignment to either treatment arm; Treatment 1 (Single Channel) and Treatment 2 (Dual Channel) are the disaggregated arm indicators, each relative to Control. Rows report the interaction of pooled and disaggregated treatment assignment with a binary indicator for (winsorized) baseline expected time to sale, testing whether the treatment effect varies along this dimension. Standard errors, clustered at the patwar-circle (PC) level, are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.10.

3.19 K-Means Clustering

We cluster on all 18 pre-specified dimensions (median-split to binary, per the pre-analysis plan’s own footnote); the eighteenth, expected time to sale, was added 22 Jul 2026 (see *Departures from Pre-Specified Analysis* above – it had not previously been operationalised anywhere in the codebase). We use $k = 3$ clusters (the pre-analysis plan does not specify a number). Because k-means cluster labels are otherwise arbitrary, the three clusters are relabelled by a simple resource-profile score (the row-sum of the 17 socioeconomic-capacity binary dimensions, with the food-insecurity indicator inverted since it is coded in the opposite direction to every other dimension) so that Cluster 1 always denotes the lowest-scoring group and Cluster 3 the highest, regardless of how the algorithm numbers them. Expected time to sale is deliberately excluded from the resource score itself – it reflects survey timing relative to the farmer’s own sale, not a socioeconomic capacity – but is still one of the 18 inputs to the clustering.

Heterogeneity by K-Means Cluster (k=3): Primary Outcomes 2024

	(1) ln(Sales _2024)	(2) ln(Quantity _2024)	(3) ln(Costs _2024)	(4) ln(Price _2024)	(5) ln(Profit _2024)
Treatment 1	0.001 (0.227)	-0.084 (0.174)	-0.147 (0.177)	0.058 (0.066)	-0.188 (0.198)
Treatment 2	0.009 (0.195)	-0.076 (0.146)	-0.117 (0.130)	0.092* (0.053)	-0.044 (0.143)
Cluster 2 (middle)	0.126 (0.262)	0.178 (0.186)	-0.020 (0.222)	0.001 (0.086)	-0.030 (0.197)
Cluster 3 (highest resource profile)	-0.351 (0.219)	-0.500*** (0.147)	-0.689*** (0.149)	0.149** (0.068)	-0.942*** (0.257)
Cluster 2 x T1	-0.291 (0.311)	-0.371 (0.247)	-0.322 (0.277)	0.091 (0.094)	-0.008 (0.246)
Cluster 2 x T2	0.001 (0.280)	0.065 (0.205)	0.129 (0.241)	-0.074 (0.108)	0.113 (0.203)
Cluster 3 x T1	0.132 (0.262)	0.286 (0.199)	0.143 (0.214)	-0.122 (0.084)	0.427 (0.275)
Cluster 3 x T2	0.030 (0.299)	0.311 (0.200)	0.222 (0.180)	-0.228** (0.106)	0.053 (0.380)
Observations	966	966	966	966	966

Table 154: K-Means Cluster Profile ($k = 3$)

	Cluster 1 (lowest resource)	Cluster 2 (middle)	Cluster 3 (highest resource)
High education	24.5%	74.6%	55.3%
High numeracy	13.2%	14.9%	23.1%
High cultivated area	13.8%	14.5%	94.2%
High agricultural empowerment	41.7%	71.5%	43.0%
High household empowerment	39.9%	74.1%	47.8%
High assets	13.8%	52.2%	77.4%
Smartphone access	46.3%	77.6%	87.1%
High agricultural income	17.8%	15.8%	94.4%
High non-agricultural income	46.9%	27.6%	37.9%
High trust	46.3%	68.9%	46.4%
High food insecurity	71.8%	15.8%	38.3%
Best price received, 2023	79.8%	45.6%	78.4%
Additional services received, 2023	4.0%	6.1%	4.9%
Understanding with buyer	83.4%	78.9%	80.8%
Buyer services at cultivation	2.5%	1.8%	1.0%
High agents considered (connectedness)	46.6%	23.2%	52.9%
High market awareness	12.3%	26.8%	31.1%
High time to sale	55.5%	37.7%	49.8%
Resource score (mean, range 0–17; excludes time to sale)	5.61	7.58	9.17
Farmers	326	228	412

Heterogeneity by K-Means Cluster (k=3): Harvest Costs, Additional Services, Credit Constraint (Expectation 2025)

	(1) ln(Harvest Cost Farmer '24)	(2) ln(Harvest Cost Buyer '24)	(3) Harvest Cost Share '24	(4) Additional Services 2024	(5) Credit Constraint (Expectation 2025)
Treatment 1	-0.254 (0.257)	1.774** (0.545)	-0.102*** (0.030)	0.200*** (0.045)	0.128*** (0.049)
Treatment 2	-0.123 (0.307)	0.404 (0.396)	-0.043 (0.031)	0.029 (0.037)	0.035 (0.038)
Cluster 2 (middle)	-0.301 (0.316)	2.071*** (0.490)	-0.173*** (0.046)	0.166*** (0.038)	0.008 (0.025)
Cluster 3 (highest resource profile)	0.932*** (0.178)	0.824** (0.346)	-0.052*** (0.014)	0.100** (0.041)	0.001 (0.026)
Cluster 2 x T1	-0.317 (0.583)	-3.023*** (0.750)	0.142** (0.072)	-0.271*** (0.062)	0.028 (0.072)
Cluster 2 x T2	0.539 (0.381)	-2.611*** (0.499)	0.194*** (0.046)	-0.175*** (0.039)	0.009 (0.046)
Cluster 3 x T1	-0.064 (0.320)	-0.358 (0.669)	0.024 (0.028)	-0.108 (0.066)	-0.033 (0.054)
Cluster 3 x T2	0.494* (0.260)	-0.949** (0.402)	0.089*** (0.028)	-0.104** (0.046)	0.000 (0.050)
Observations	966	966	966	966	966

4 Attrition

The pre-analysis plan (Section 4) commits to testing for differential attrition between treatment arms, by repeating the baseline balance-test specification with a round-specific attrition dummy as the outcome. Table 4 reports this test for each of the three post-baseline survey rounds (in-person endline, phone reminder, phone follow-up), for both the pooled Treatment indicator and the disaggregated Treatment 1/Treatment 2 indicators.

	(1) In-person Follow-up Harvest 2024	(2) In-person Follow-up Harvest 2024	(3) Phone Reminder Sowing 2025	(4) Phone Reminder Sowing 2025	(5) Phone Follow-up Harvest 2025	(6) Phone Follow-up Harvest 2025
Treatment	-0.009 (0.018)		-0.007 (0.031)		0.021 (0.021)	
Single		-0.014 (0.019)		0.016 (0.037)		0.019 (0.026)
Dual		-0.003 (0.022)		-0.029 (0.036)		0.022 (0.024)
Mean	0.06	0.06	0.29	0.29	0.18	0.18
Observations	1058	1058	1058	1058	1058	1058

5 Robustness

The pre-analysis plan (Section 5) commits to two robustness checks repeating the main ITT estimation: (i) village fixed effects in place of randomisation-strata fixed effects, and (ii) post-double-selection with LASSO using the baseline covariates listed in the baseline PAP.

5.1 Village Fixed Effects

This check is not estimable given the realised randomisation design. Randomisation was conducted at the patwar circle (PC) level, and every sampled village sits entirely within a single PC; as a result, treatment assignment has zero within-village variation (confirmed directly in the data: 0 of 98 villages show any variation in Treated, Treatment 1, or Treatment 2). Village fixed effects are therefore perfectly collinear with treatment, and the treatment coefficient cannot be identified under this specification. This is a mismatch between the pre-analysis plan (drafted before the final randomisation level was confirmed) and the implemented design, not a computational issue, and is logged as a PAP deviation rather than silently omitted.

5.2 Post-Double-Selection LASSO

Candidate controls are the respondent and household baseline variables listed in the baseline pre-analysis plan (Section 3.2): male, age, education, numeracy, experience, empowered_1, empowered_2, trust, household size, assets, cultivated area, and food insecurity. Village fixed effects, also named as a candidate control in the pre-analysis plan, are excluded here for the same reason given above: treatment has zero within-village variation, so including village dummies as LASSO candidates would let the treatment-equation selection step pick enough of them to reproduce the same non-identification problem.

[illegible]